HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MASTER THESIS

Vietnamese Voice Conversion using Deep Learning Technique

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Abstract

Voice transformation (VT) aims to change one or more aspects of a speech signal while preserving linguistic information. A subset of VT, Voice conversion (VC) aims at altering a person's voice to make it sound similar to the voice of another person while preserving linguistic content. Studies started from 1980s, the original motivation of voice conversion could be simply novelty and curiosity, the technological advancements from statistical modeling to deep learning have made a major impact on many real-life applications, such as entertainment, healthcare, education...

Nowadays, with the remarkable advancements recently in deep learning technologies, voice conversion models are increasingly achieving high quality and moving towards one-shot models, meaning that we only need a single sample to capture the voice of any given person. However, existing methods suffer from a dilemma between content intelligibility and speaker similarity for unseen data; i.e., methods with higher intelligibility usually have a lower speaker similarity, while methods with higher speaker similarity usually require plenty of target speaker voice data to achieve high intelligibility [1]. Recently, the Phoneme Hallucinator model has been proposed, utilizing hallucination mechanisms to enrich data from one-shot samples. This model largely addresses the aforementioned issue and yields promising results. However, the models mentioned above are trained on non-tonal datasets such as English. For tonal languages like Vietnamese, these models yield poor results and often lose the tonal characteristics of the original language. Moreover, for Vietnamese, there are lack of large multi-speaker datasets such as VTCK [2], LibriSpeech [3] for English. These things lead to the poor quality of voice conversion models for Vietnamese.

This thesis develops a voice conversion model for Vietnamese based on the Phoneme Hallucinator [4] model with 2 adoptions: (1) Add a Text2SSL module to get more context information before performing the KNN algorithm, (2) To create a more diverse dataset we apply spectrogram-resize (SR) based data augmentation idea from Free-VC model [5] which distorts speaker information without changing content information to generate more ”speakers”.

Keywords: Speech Synthesis, Voice Transformation, Voice Conversion, Deep Learning, One-Shot Learning

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ACRONYMS

Notation	Description
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
BLSTM	Bidirectional Long Short-Term Memory
CER	Character Error Rate
CNN	Convolutional Neural Network
CNNs	Convolutional Neural Networks
DNN	Deep Neural Network
E2E	End To End
GAN	Generative Adversarial Network
HMM	Hidden Markov Model
KNN	K-Nearest Neighbor
LSTM	Long Short Term Memory
MOS	Mean Opinion Score
NLP	Natural Language Processing
RNN	Recurrent Neural Network
SOTA	State Of The Art
SPSS	Statistical Parametric Speech Synthesis
SR	Spectrogram Resizing
SUS	Semantically Unpredictable Sentences
TTS	Text To Speech
VAE	Variational Auto Encoder
VC	Voice Conversion
VT	Voice Transformation
WER	Word Error Rate

CHAPTER 1. INTRODUCTION

1.1 Introduction to voice conversion

Voice conversion (VC) is a significant aspect of artificial intelligence. It is the study of how to convert one's voice to sound like that of another without changing the linguistic content. Voice conversion belongs to a general technical field of speech synthesis, which converts text to speech or changes the properties of speech, for example, voice identity, emotion, and accents [1]. The the really difficult problem involved in the the artificial production of speech sounds is not the making of a device which shall produce speech, but in the manipulation of the apparatus. As voice conversion is focused on the manipulation of voice identity in speech, it represents one of the challenging research problems in speech processing. Figure 1.1 shows an illustration of voice conversion which alters the pitch, intonation, and rhythm of voice A with corresponding characteristics of voice B.

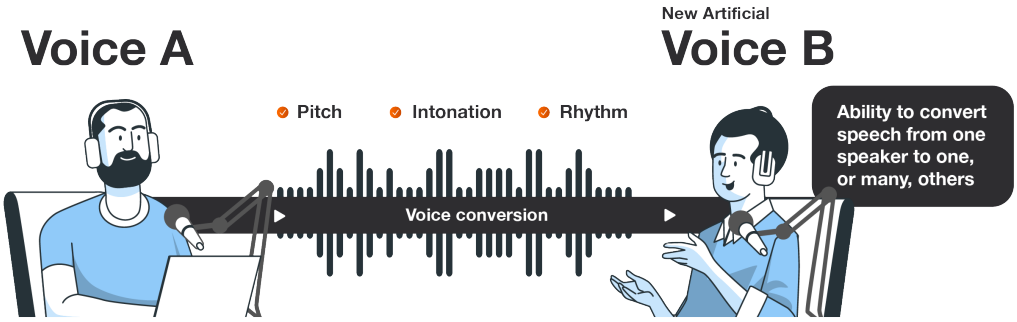


Figure 1.1: The illustration of voice conversion

There has been a continuous effort in quest for effective manipulation of speech properties since the debut of computer-based speech synthesis in the 1980s. While the original motivation of voice conversion could be simply novelty and curiosity, the technological advancements from statistical modeling to deep learning have made a major impact on many real-life applications, and benefited the consumers, such as personalized speech synthesis communication aids for the speech impaired speaker de-identification voice mimicry and disguise and voice dubbing for movies. In the next section we will show more about applications of voice conversion.

1.2 Applications of voice conversion

Voice conversion technology is a fascinating and rapidly evolving field that intersects with various disciplines, including digital signal processing, machine learning, linguistics, and psychology. At its core, voice conversion aims to alter a person's voice to sound like another, maintaining the linguistic content while changing the acoustic properties. This technology is not just about altering pitch or speed; it's about capturing and transferring the unique vocal identity of one individual to another, which involves a complex interplay of various speech characteristics.

The applications of voice conversion are as diverse as they are impactful, stretching across entertainment, personalization, accessibility, and security. These applications leverage the core ability of voice conversion to transform voice characteristics while preserving the message's intelligibility and naturalness. As the technology matures, its integration into our daily lives becomes more seamless and its benefits more apparent, touching on aspects of personal identity, privacy, and digital interaction. In this thesis, we present four common applications of voice conversion as the following:

1.2.1 Enhancing User Experiences

Voice conversion technology stands as a pivotal innovation in the entertainment industry, particularly in the realms of cinema, animation, and gaming. This technology enables actors to alter their vocal attributes to align with various characters, fostering an immersive narrative experience. In animation and gaming, where characters may possess fantastical attributes, voice conversion allows for the creation of voices that match these unique characteristics, thus enhancing the believability and engagement of the narrative.

Researchers have explored the implications of voice conversion in entertainment, noting its capacity to broaden the range of roles an actor can portray, thereby augmenting their versatility and the diversity of the content produced [6]. For instance, an actor with a distinctively deep voice can employ voice conversion to portray a character requiring a higher pitch, expanding their repertoire and the realism of the character portrayal. Moreover, voice conversion facilitates the localization of content, enabling producers to adapt voices to different languages or regional accents, thereby expanding their audience reach. This aspect of voice conversion is particularly beneficial in global markets, where cultural and linguis-

tic diversity necessitates content adaptation for regional consumption [6].

The technology's impact extends to enhancing user experience in interactive media. In video games, for instance, players can customize character voices, allowing for a more personalized gaming experience. This level of customization not only increases player engagement but also contributes to the narrative depth of the game [7]. Despite its benefits, the use of voice conversion in entertainment is not without challenges. Concerns regarding authenticity and the potential for misuse in misrepresenting individuals highlight the need for ethical considerations and regulatory frameworks in this domain [6].

1.2.2 Voice Anonymization

Nowadays, we see more opportunities to use speech dialogue with machines and E-commerce such as online shopping and online banking. With the value of speech as personal information drawing attention, privacy protection and security aspects surrounding the use and share of speech data is becoming a concerning element. Anonymization is a well-known solution for privacy protection, it refers to the goal of suppressing personally identifiable information in the speech signal, leaving other attributes intact such as linguistic content.

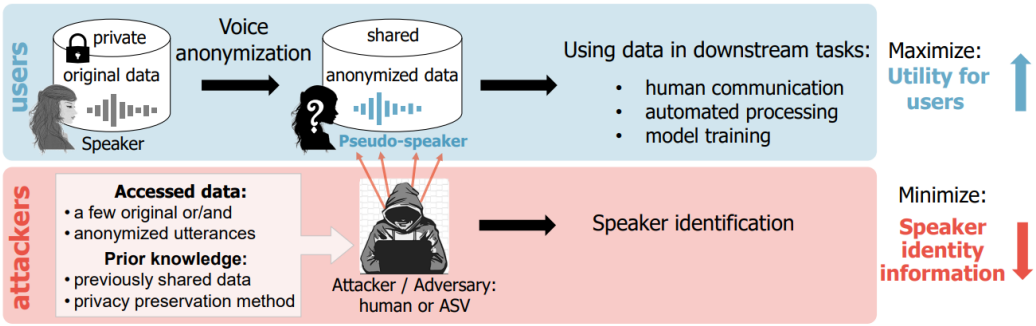


Figure 1.2: Example of a privacy preservation scenario between users and attackers in the case where speaker identity is considered as personal information to be protected [8]

Voice conversion technology's application in telecommunications, particularly for voice anonymization, presents a significant advancement in enhancing privacy and security. By modifying voice characteristics, individuals can communicate without revealing their true voice identity, which is crucial in scenarios like witness protection or confidential business communications [6]. Figure 1.2 presents an example of a privacy preservation scenario between users and attack-

ers in the case where speaker identity is considered as personal information to be protected. The technology operates by altering voice parameters such as pitch, timbre, and intonation, effectively disguising the speaker's identity while maintaining the intelligibility of the spoken content. This capability is invaluable in protecting the anonymity of individuals in sensitive situations, offering them a layer of security that is difficult to breach with conventional methods [9].

Furthermore, voice anonymization can serve in customer service environments, where representatives may choose to mask their voices to prevent recognition or harassment outside of work. This application underscores the technology's role in safeguarding personal privacy in professional settings [10]. However, the implementation of voice conversion for anonymization purposes necessitates a careful balance between privacy protection and potential misuse, such as identity deception. As such, ongoing research aims to refine these technologies to enhance security features while minimizing opportunities for unethical use [11].

1.2.3 Accessibility for Individuals with Speech Impairments

Voice conversion technology holds transformative potential for individuals with speech impairments, offering them new avenues for communication. By converting text to speech or modifying existing speech, individuals can communicate in a manner that more closely resembles their natural voice, enhancing their ability to express themselves and interact with others. This application not only improves the quality of life for individuals with speech impairments but also fosters inclusivity, allowing for more effective communication in various social and professional contexts. The ongoing refinement of voice conversion technology in this domain aims to provide more personalized, natural-sounding voice options, catering to the diverse needs and preferences of users.

Moreover, the technology can adapt to the specific needs and preferences of the user, allowing for customization that accounts for the nuances of individual speech patterns. This level of personalization not only improves the usability of assistive devices but also contributes to the psychological well-being of the user by preserving their vocal identity [12]. As the technology advances, ensuring user-centric design and addressing potential ethical concerns will be paramount in maximizing the benefits of voice conversion for individuals with speech impairities, contributing to a more inclusive and communicative society.

1.2.4 Language Learning

The field of language learning has been notably enriched by voice conversion technology, providing learners with innovative tools to improve pronunciation and accent. Voice conversion can modify a learner's recorded speech to align with native pronunciation, offering immediate and personalized feedback. This application not only accelerates the learning process but also enhances the learner's confidence and comprehension [11].

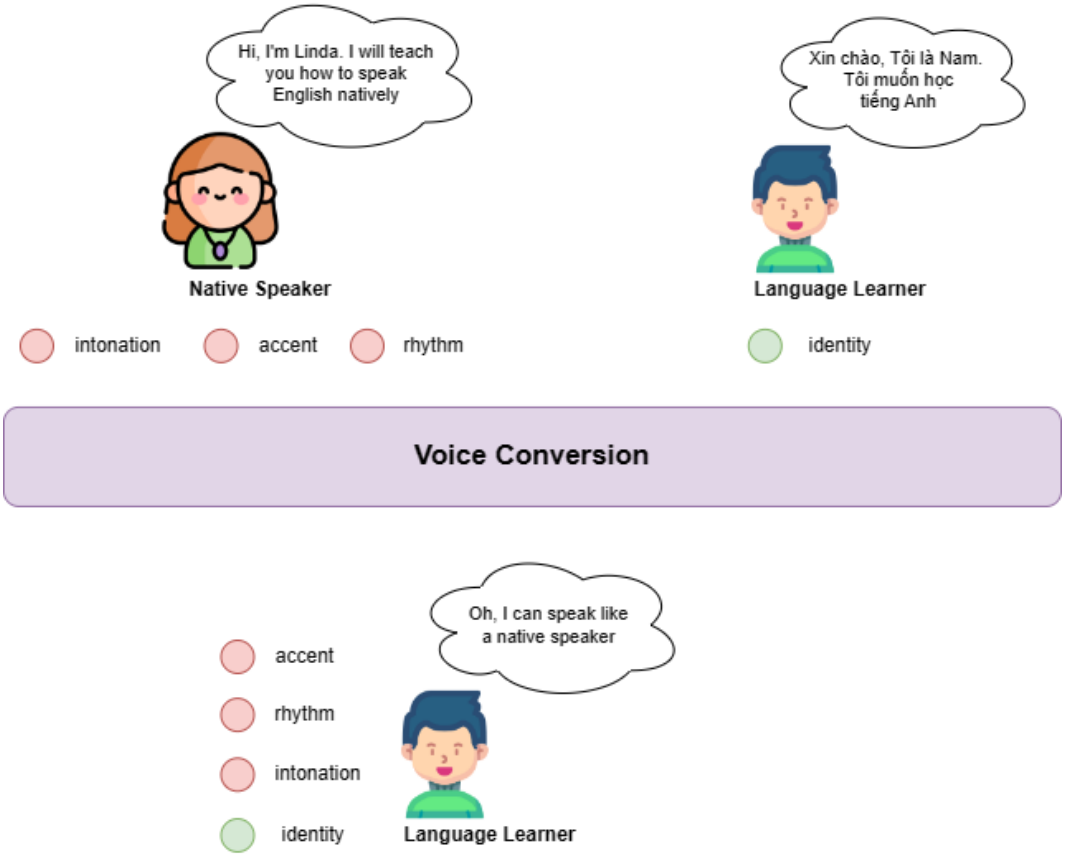


Figure 1.3: The output audio has the identity of learner with English characteristics such as accent, rhythm & intonation

In addition to pronunciation training, voice conversion can facilitate language translation and dubbing in educational materials, making content accessible to a broader audience. This utility is especially relevant in an increasingly interconnected world, where multilingual resources can significantly impact educational outreach and effectiveness. Future advancements in voice conversion for language learning will likely focus on enhancing the naturalness and accuracy of voice modification, ensuring that learners receive high-quality, realistic feedback that genuinely advances their language skills.

1.3 The Current State of Research

1.3.1 Global Research Landscape

In recent years, there has been a surge in research and development efforts worldwide focused on advancing voice conversion technology. Traditional approaches, such as Gaussian mixture model (GMM) [13] based methods, have been extensively studied and refined to achieve better performance in terms of conversion quality and efficiency. Additionally, deep learning techniques, particularly neural network-based models like variational autoencoders (VAEs) [14] and generative adversarial networks (GANs)[15], have gained prominence for their ability to capture complex nonlinear relationships in speech data, leading to significant improvements in voice conversion quality. A few exemplary models that can be mentioned include FreeVC [5] and KNN-VC [16], which have achieved audio conversion with quality approaching human-level without annotated training data. Moreover, recently, voice conversion models are increasingly achieving high quality and moving towards one-shot models, meaning that we only need a single sample to capture the voice of any given person. One of the popular models is Phoneme Hallucinator utilizing hallucination mechanisms to enrich data from one-shot samples. This model has a reasonable architecture & yields promising results for the English datasets.

Several competitions, such as the Voice Conversion Challenge (VCC) series [17], have been established to facilitate objective evaluation and comparison of different VC algorithms. The released datasets from these competitions feature a diverse range of speakers and linguistic content, enabling researchers to assess the generalization capabilities of their models across various speaking styles and languages. Furthermore, interdisciplinary collaborations between researchers from fields such as machine learning, signal processing, and linguistics have contributed to the rapid advancement of voice conversion technology. These collaborations have led to the development of novel techniques and methodologies for addressing challenging aspects of voice conversion, such as speaker variability, linguistic context preservation, and the naturalness of converted speech.

1.3.2 Research Landscape in Vietnam

With our experiments, when applying models such as FreeVC, KNN-VC, and Phoneme Hallucinator to the one-shot voice conversion problem for Vietnamese, the audio quality is relatively poor, and the tonal characteristics of the original language are lost. These phenomena have been investigated and explained below:

- While the global research community has made significant strides in improving the naturalness and intelligibility of converted voices, the focus has largely been on non-tonal languages. For example, the well-known WavLM [18] representation learning model, which is used to convert audio into feature vectors, has been trained for hundreds of thousands of hours on non-tonal data and this model serves as a crucial input encoder for voice conversion tasks. That has resulted in a wealth of knowledge and methodologies that, while effective in general, may not fully address the peculiarities of tonal languages like Vietnamese.

- Most of the existing research in Vietnam has been primarily focused on basic speech processing tasks, such as speech recognition and synthesis, with relatively fewer studies dedicated to voice conversion. Several speech processing challenges have been organized such as Zalo AI Challenge [19], VLSP [20] but there are no shared tasks for voice conversion. Furthermore, the lack of publicly available Vietnamese speech corpora and benchmark datasets such as VTCK [2], LibriSpeech [3] which poses a significant challenge for researchers in Vietnam, hindering the development and evaluation of voice conversion systems tailored to the Vietnamese language.

However, these challenges also present opportunities for future research and innovation. In this thesis, we will fill this gap by adapting the one-shot voice conversion model - Phoneme Hallucinator [4] with two modifications:

- (1) Focusing on the tonal aspects of voice conversion and cross-linguistic adaptation by proposing a new Text2SSL module that generates self-supervised learning features from the source text. The purpose of this component is to combine linguistic information from text into the one extracted by WavLM.

- (2) To create a multi-speakers dataset we apply the augmentation idea from the FreeVC model [5] to generate more ”speakers” from the original corpus.

The work being done in Vietnam not only contributes to the global body of knowledge but also serves as a pivotal point of reference for research in other tonal languages. In the following chapters, we will delve deeper into the methodologies, experiments, and proposed solutions aimed at advancing voice conversion research, with a focus on addressing the unique challenges posed by the Vietnamese language.

CHAPTER 2. VOICE CONVERSION METHODOLOGIES

In the previous chapter, we explored the foundational concepts of voice conversion (VC), examining its significance, applications, and the state of research both globally and within the Vietnamese context. Having established the relevance and identified the challenges inherent in VC, this chapter delves deeper into the methodologies that underpin this transformative technology. We trace the arc from traditional signal processing techniques to the modern incursion of deep learning paradigms, systematically unpacking the mechanics, strengths, and limitations of each methodology. In this chapter, we will go through a typical flow of a voice conversion model, then we will delve into the detailed development of voice conversion models from traditional to deep learning models. From there, we will analyze the advantages and disadvantages and establish the theoretical foundation for developing voice conversion models for Vietnamese.

2.1 Typical flow of voice conversion

The goal of voice conversion is to modify a source speaker's voice to sound as if it is produced by a target speaker. In other words, a voice conversion system only modifies the speaker-dependent characteristics of speech, such as formants, fundamental frequency (F0), intonation, intensity and duration, while carrying over the speakes-independent speech content [1].

A typical voice conversion pipeline includes a speech analysis, mapping, and reconstruction modules as illustrated in Figure 2.1, which is referred to as the analysis-mapping reconstruction pipeline. The speech analyzer decomposes the speech signals of a source speaker into features that represent supra-segmental and segmental information, and the mapping module changes them towards the target speaker, finally, the reconstruction module re-synthesizes time-domain speech signals. The mapping module has taken center stage in many of the studies. These techniques can be categorized in different ways, for example, based on the type of statistical modeling technique - parametric vs non-parametric, the scope of optimization - frame level vs utterance level, and the methods for disentangling content representations - text-based vs text-free. We will discuss these techniques in more detail in the next sections.

The core module of a voice conversion system performs the conversion function. Let's denote the source and target speech signals as X and Y respectively. As

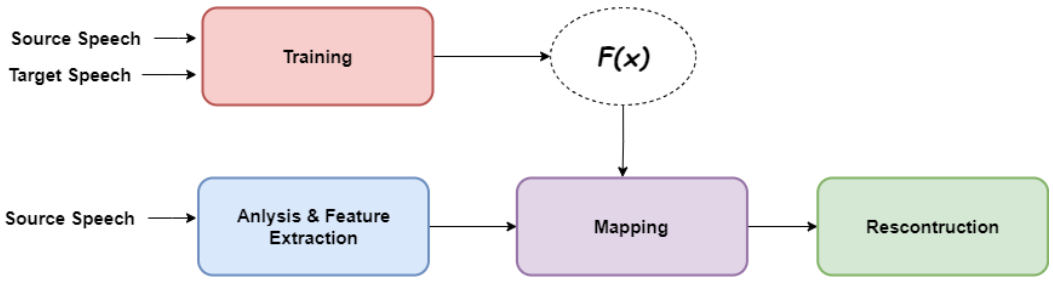


Figure 2.1: The typical flow of a voice conversion system

will be discussed later, voice conversion is typically applied to some intermediate representation of speech, or speech feature, that characterizes a speech frame. Let's denote the source and target speech features as x and y. The conversion function can be formulated as follows,

$$y = F(x) \tag{2.1}$$

where $F(x)$ is also called the frame-wise mapping function in the rest of this thesis. As illustrated in Figure 2.1, a typical voice conversion framework is implemented in three steps: 1) speech analysis, 2) feature mapping, and 3) speech reconstruction, that we call the analysis-mapping-reconstruction pipeline.

2.1.1 Speech Analysis and Reconstruction

The goal of speech analysis is to decompose speech signals into some form of intermediate representation for effective manipulation or modification with respect to the acoustic properties of speech. There have been many useful intermediate representation techniques that were initially studied for speech communication, and speech synthesis. They become handy for voice conversion. In general, the techniques can be categorized into model-based representations, and signal based representations.

In model-based representation, we assume that speech signal is generated according to a underlying physical model, such as source-filter model, and express a frame of speech signal as a set of model parameters. By modifying the parameters, we manipulate the input speech. In signal based representation, we don't assume any models, but rather represent speech as a composition of controllable elements in time domain or frequency domain. Let's denote the intermediate representation for source speaker as z, speech analysis can be described by a function

$$z = A(x) \tag{2.2}$$

Speech reconstruction can be seen as an inverse function of the speech analysis, that operates on the modified parameters and generates an audible speech signal. It works with speech analysis in tandem. For example, A vocoder [21] is used to express a speech frame with a set of controllable parameters that can be converted back into a speech waveform. A Griffin-Lim algorithm is used to reconstruct a speech signal from a modified short-time Fourier transform after amplitude modification [22]. As the output speech quality is affected by the speech reconstruction process, speech reconstruction is also one of the important topics in voice conversion research. Let's denote the reconstruction function as R, voice conversion can be described by a composition of three functions,

$$Y = (R \circ F \circ A)(x) = C(x) \tag{2.3}$$

that represents the typical flow of a voice conversion system as a 3-step pipeline. As the mapping is applied frame-by frame, the number of converted speech features y is the same as that of the source speech features x if speech duration is not modified in the process.

2.1.2 Feature Extraction

Typically, audio is split into frames and each speech frame is represented by a feature vector. Feature extraction aims to derive low-dimensional spectral representations from the high-dimensional raw waveform. Generally speaking, the spectral features are be able to represent the speaker individuality well. The feature not only fit the spectral envelope well, but also be able to be converted back to spectral envelope. They should have good interpolation properties that allow for flexible modification. In this way, high dimension magnitude spectrum is transformed to lower dimension feature representation. The commonly used speech features include Mel frequency cepstral coefficients (MFCC) [23], and Mel-Spectrogram [24] as visualized in Figure 2.2

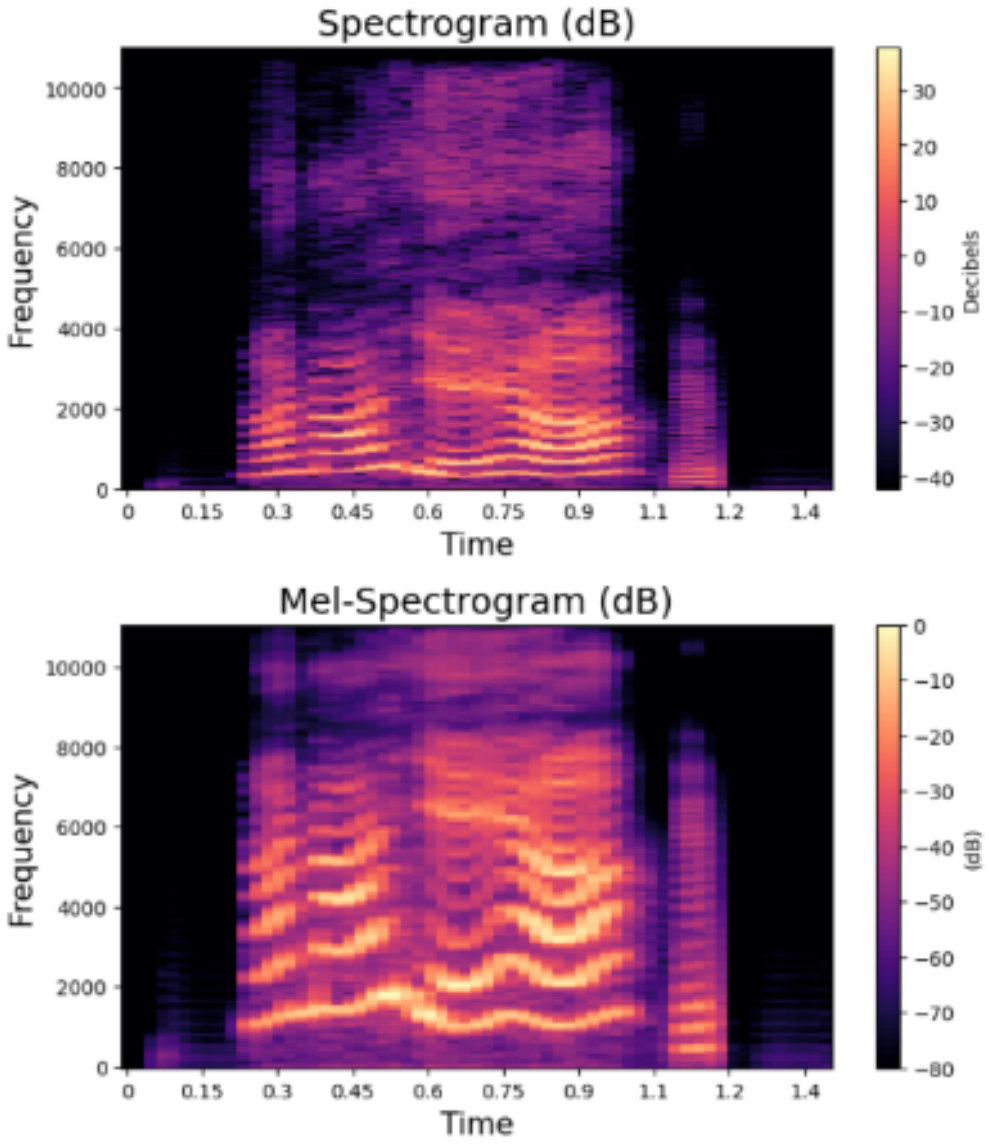


Figure 2.2: An example of spectrogram, and mel-spectrogram [24]

A spectrogram is a visual representation of a signal's frequency spectrum as it changes over time [24]. In the field of speech, spectrograms are often used to analyze and change speech data. The spectrogram represents the frequency content of a spoken signal over time. The x-axis indicates time, while the y-axis indicates frequency. At each location on the spectrogram, the color intensity corresponds to the amplitude or strength of the corresponding frequency component. Spectrograms are useful for judging how speech sounds, figuring out what phonetic qualities they have, and learning about how speech sounds. They are also used in voice conversion to synthesize new speech and analyze the input audio. Based on the idea that the human ear is more sensitive to some frequencies than others, this property tries to compress the representation of speech in the higher frequency

domain. The mel scale is an experimental function that shows how sensitive the human ear is to different frequencies. The mel-spectrogram, which is based on the auditory-based mel-frequency scale, gives more frequency resolution than the spectrogram [24].

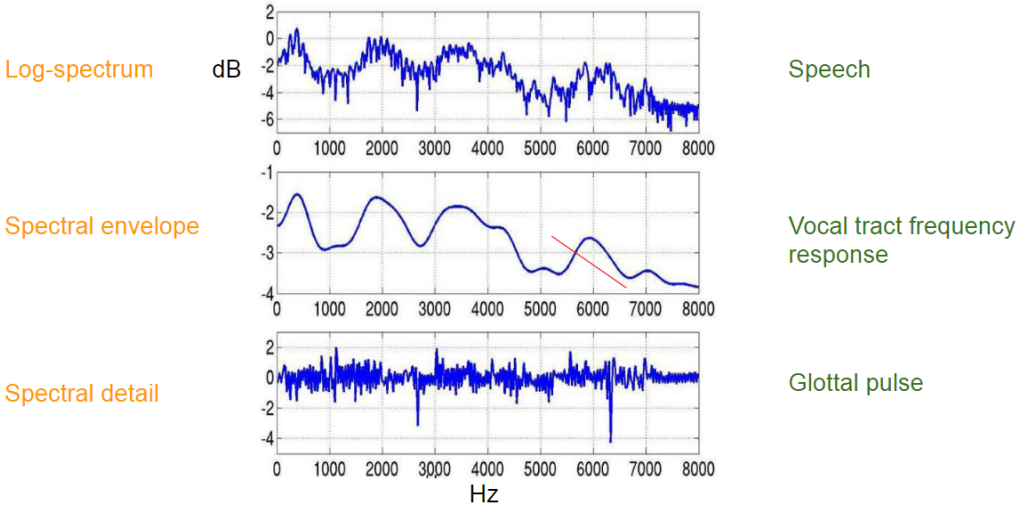


Figure 2.3: Mel Frequency Cepstrum Coefficient (MFCC)

Similar to mel-spectrograms, MFCC uses a mel scale to approximate human perception of sound. However, MFCC uses fewer coefficients (typically 13-20) compared to the number of frequency bands in a mel-spectrogram (usually 32-80). This makes it more computationally efficient. It will be suitable for audio analysis due to its compactness and robustness to noise.

2.2 Traditional Voice Conversion Model

Voice conversion (VC) has a rich history, with traditional approaches laying the groundwork for the development of modern techniques. In this section, we explore traditional methodologies employed in voice conversion, including Non-Negative Matrix Factorization (NMF) [25], statistical parametric methods, Gaussian Mixture Model (GMM)-based techniques [13], and Hidden Markov Model (HMM)-based approaches [26]. Most of the traditional voice conversion techniques assume availability of parallel training data. In other words, the mapping function is trained on paired utterances of the same linguistic content spoken by source and target speaker.

2.2.1 Non-Negative Matrix Factorization (NMF)

Non-Negative Matrix Factorization (NMF) [25] offers an alternative perspective on voice conversion, decomposing speech data into non-negative basis components. NMF represents speech features as a linear combination of basis vectors and corresponding coefficients, ensuring that both the basis and coefficients are non-negative. In NMF-based voice conversion, the basis matrix learned from the source speaker's speech features is transformed to approximate the basis matrix of the target speaker. This transformation facilitates the conversion of speech features while preserving linguistic content.

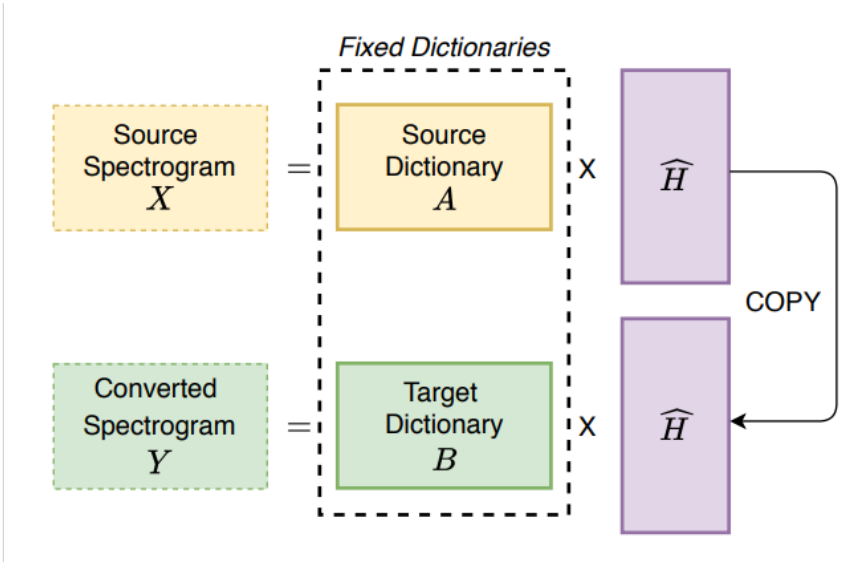


Figure 2.4: Non-Negative Matrix Factorization (NMF) - An example of non-parametric technique in voice conversion [25]

2.2.2 Gaussian Mixture Model (GMM)-Based Techniques

Statistical parametric methods form the foundation of voice conversion research, relying on statistical modeling techniques to transform speech features from one speaker to another. These methods operate on a set of acoustic features extracted from speech signals, such as Mel-Frequency Cepstral Coefficients (MFCCs) or Line Spectral Frequencies (LSFs) [24].

One common statistical parametric method is Gaussian Mixture Model (GMM) [13]. GMM-based techniques are widely used in voice conversion due to their simplicity and effectiveness. These methods assume that the joint distribution of source and target speakers' speech features can be modeled as a mixture of

Gaussian distributions. In GMM-based voice conversion, the source speaker's speech features are first mapped to a high-dimensional space using a non-linear transformation, such as Gaussianization or Maximum Likelihood Linear Transformations (MLLT). Then, a GMM is trained to model the relationship between the transformed source features and the target speaker's features.

During conversion, the GMM parameters are used to estimate the target speaker's features from the source speaker's features, effectively transforming the voice while preserving linguistic content. However, GMM-based techniques may struggle with capturing non-linear relationships and producing natural-sounding speech, particularly in cases of significant speaker mismatch.

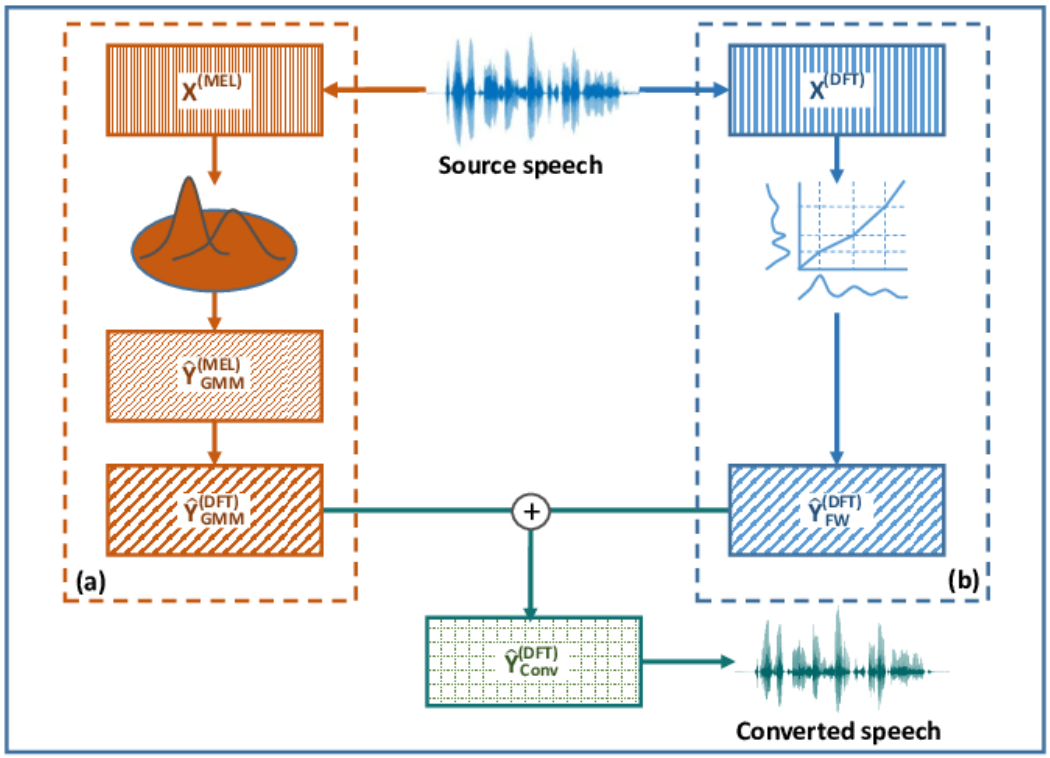


Figure 2.5: GMM-based voice conversion [13]

2.2.3 Hidden Markov Model (HMM)-Based Approaches

Hidden Markov Model (HMM)-based approaches are another class of traditional voice conversion techniques, leveraging HMMs to model the temporal dynamics of speech features. In HMM-based voice conversion, source and target speakers' speech features are represented as sequences of states in the HMM. The HMM parameters, including transition probabilities and emission probabilities, are estimated from training data using algorithms such as Expectation-Maximization

(EM) or Baum-Welch [26].

During conversion, the source speaker's speech features are aligned with the target speaker's HMM, and the most likely sequence of target states is inferred using algorithms such as Viterbi [27] decoding. This sequence is then used to generate converted speech features corresponding to the target speaker's voice.

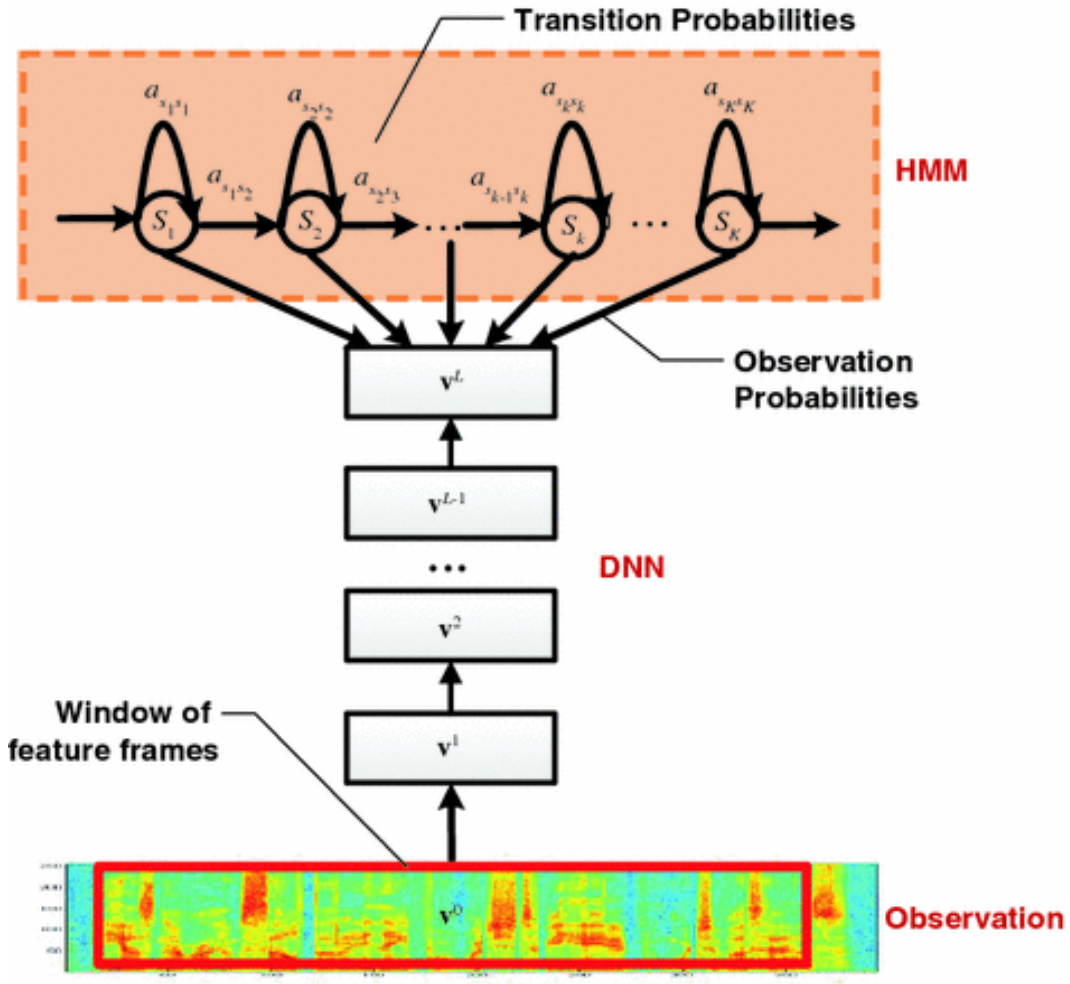


Figure 2.6: HMM-based voice conversion: source and target speakers' speech features are represented as sequences of states [26]

In general, while traditional methods have paved the way for advancements in voice conversion, they often face limitations in capturing the intricacies of natural speech and adapting to diverse speaker characteristics. As a result, there has been a shift towards modern approaches, such as deep learning-based methodologies, which aim to address these challenges and achieve improved performance in voice conversion tasks.

2.3 Recurrent Neural Networks (RNNs) in VC

Recurrent Neural Networks (RNNs) [28] are well-suited for sequential data processing tasks, making them suitable for modeling temporal dependencies in speech signals. In voice conversion, RNNs are utilized to capture long-range dependencies and temporal dynamics inherent in speech features. Long Short-Term Memory (LSTM) [29] and Gated Recurrent Unit (GRU) **gru** variants of RNNs are commonly employed due to their ability to mitigate the vanishing gradient problem and retain memory over long sequences. RNN-based VC models operate on sequential representations of speech, such as mel-spectrograms or waveform samples. The network is trained to predict the target speaker's features from the source speaker's features, leveraging parallel data for supervised learning. By capturing temporal dependencies, RNNs can effectively model speech dynamics and generate natural-sounding converted speech. The figure 2.7 shows a RNN-based voice conversion model called DBLSTM-RNNs [30], which was created by stacking multiple Bidirectional Long Short-Term Memory (BLSTM) [29] hidden layers. It allows the best of long-range context in both forward and backward directions.

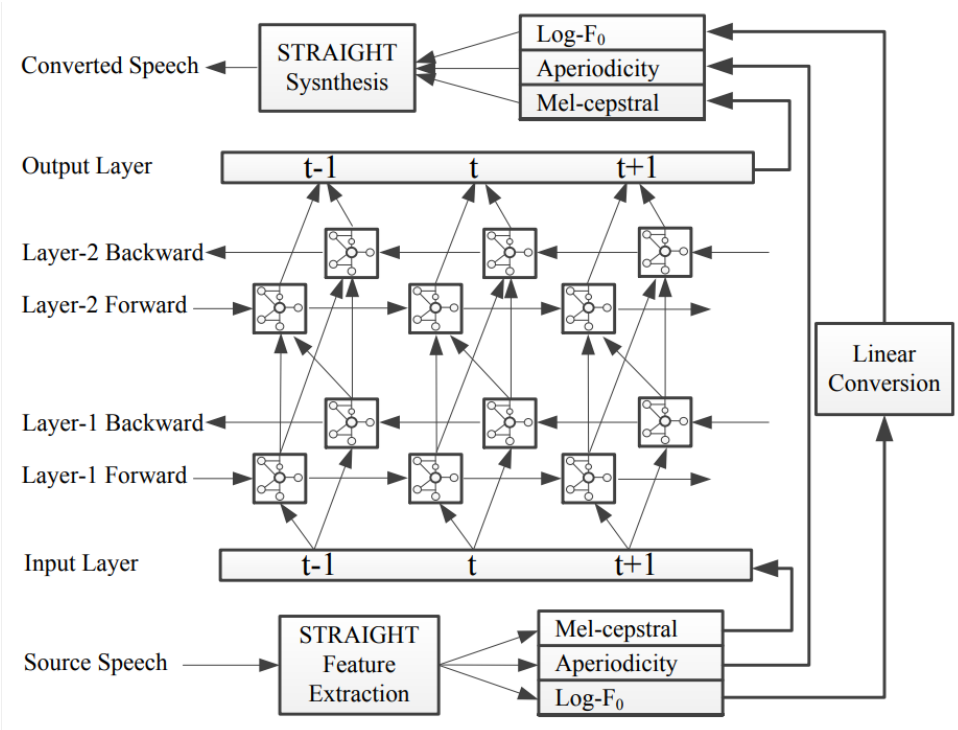


Figure 2.7: Fig. 4. The DBLSTM-RNN based voice conversion architecture [30]

2.4 Variational Autoencoders (VAEs) for VC

Variational Autoencoders (VAEs) [14] offer a probabilistic approach to learning latent representations of data, enabling generative modeling and data synthesis. In voice conversion, VAEs are utilized to learn disentangled representations of speech features, separating speaker identity from linguistic content. By imposing a Gaussian prior on the latent space, VAEs encourage smooth interpolation between speakers and facilitate diverse and controllable speech synthesis.

In VAE-based VC models, the encoder network maps the input speech features to a probability distribution in the latent space, while the decoder network reconstructs the target speaker's features from samples drawn from this distribution. By optimizing the reconstruction loss and the Kullback-Leibler [31] divergence between the learned distribution and the prior, VAEs learn to disentangle speaker identity from other speech attributes.

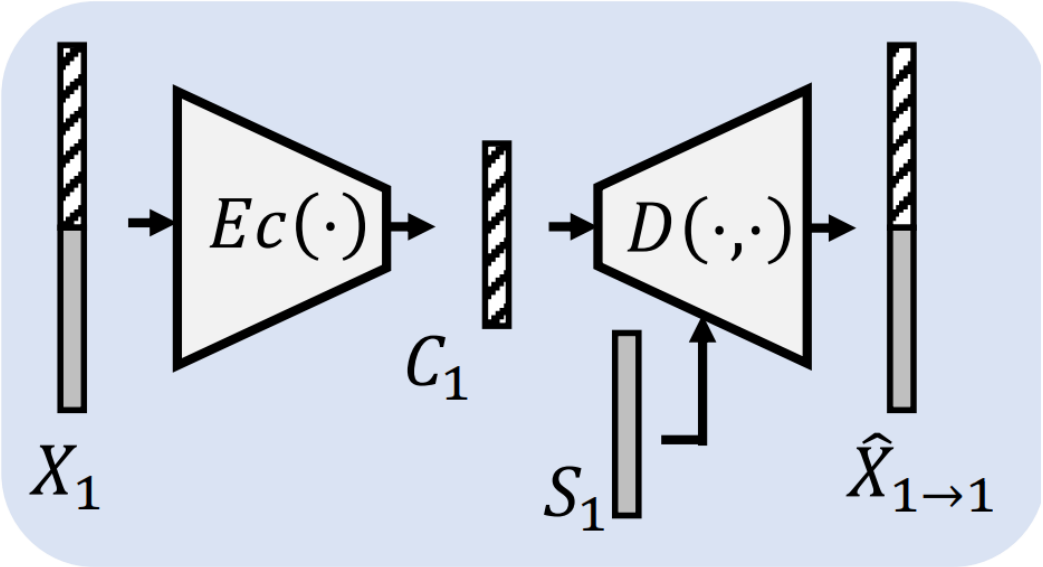


Figure 2.8: AutoVC: An example of disentangling content information and speaker information (Each speech segment contains two types of information: the speaker information - solid and content information - striped) [32]

AutoVC [32] is a notable voice conversion model using Variational Autoencoder (VAE) [14] architecture introduced by Qian et al. in 2019. AutoVC is a zero-shot voice conversion model, meaning it can perform voice style transfer without paired data. By leveraging the VAE framework, AutoVC learns a latent space representation of speech features, which enables it to generate target speaker features from source speaker input without direct supervision. This approach results

in natural and expressive voice conversions while preserving the linguistic content of the input speech. AutoVC represents a significant advancement in voice conversion research, offering a practical solution for voice style transfer tasks with minimal data requirements and high conversion quality.

2.5 Generative Adversarial Networks (GANs) in VC

Generative Adversarial Networks (GANs) [15] have gained prominence for their ability to generate realistic and high-fidelity samples in various domains, including image and speech synthesis. In voice conversion, GANs are employed to learn a mapping between source and target speaker features by simultaneously training a generator and a discriminator network in an adversarial fashion. The generator network in a GAN-based VC model learns to generate converted speech features that are indistinguishable from those of the target speaker, while the discriminator network distinguishes between genuine and synthesized samples. Through adversarial training, the generator improves its ability to produce convincing converted speech, while the discriminator learns to provide accurate feedback to guide the generator's learning process.

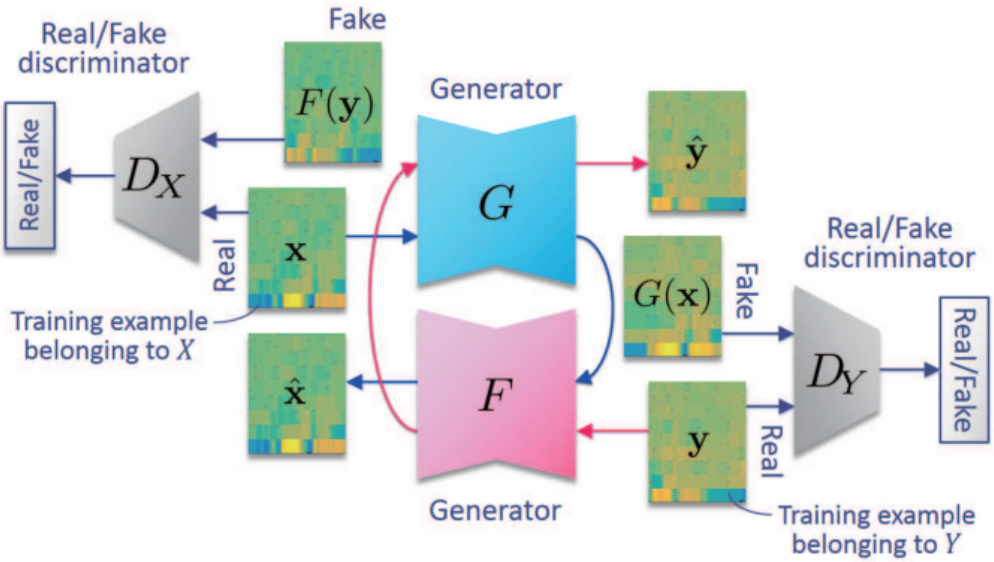


Figure 2.9: The training flow of CycleGAN

One popular GAN-based VC architecture is the CycleGAN [33] framework adapted for speech signals, which employs cycle-consistency to enforce bi-directional conversion between source and target speakers. By leveraging adversarial training, GAN-based VC models can produce high-quality converted speech with im-

proved naturalness and speaker similarity. But the biggest limitation of CycleGAN is that it only learns one-to-one-mappings. So, another model called StarGAN-VC [21] has been proposed to tackle this limitation which allows non-parallel many-to-many VC across different attribute domains using a single generator network. Moreover, unlike CycleGAN-VC, StarGAN-VC does not require any information about the attribute of the input speech at test time.

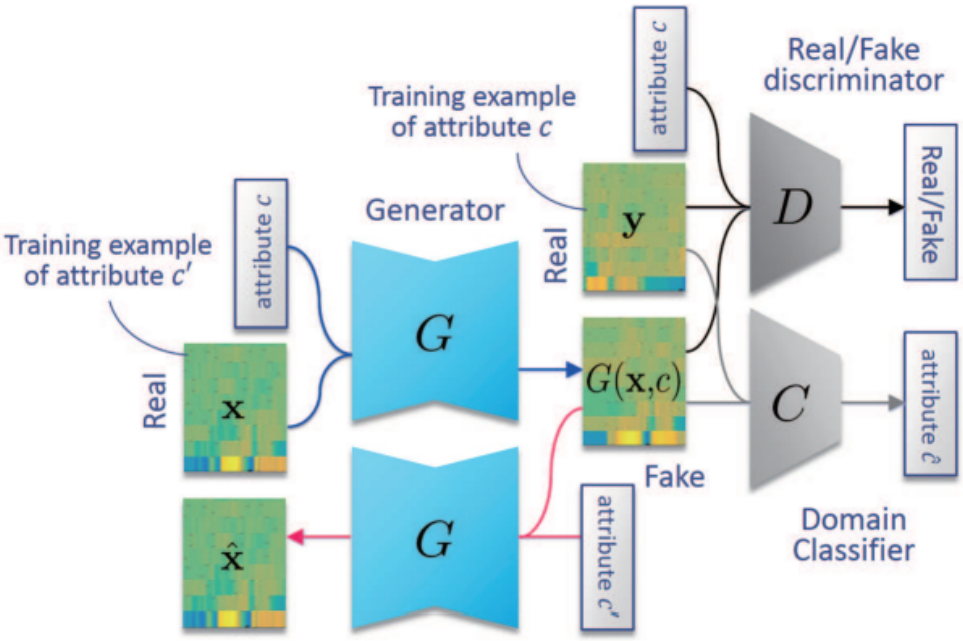


Figure 2.10: The training flow of StarGAN [21]

2.5.1 One-shot Voice Conversion

While the aforementioned models such as AutoVC [32], CycleGAN [33] can generate high-quality output speech, they typically require several minutes to hours of training examples to produce reasonably realistic-sounding speech. This hinders their practical applicability since such a large amount of data is not always readily available. Consequently, there has been a growing interest in developing models that can achieve voice conversion with only 3-10 seconds of target speech. Several one-shot voice conversion models worth mentioning include VQMIVC, BNE-PPG-VC, and the recently prominent FreeVC [5]. These models exemplify significant advancements in the field of voice conversion, demonstrating the ability to transform a speaker's voice using only a single audio sample of the target speaker. VQMIVC utilizes a variational autoencoder framework to capture and convert vocal characteristics, while BNE-PPG-VC employs bottle-

neck feature extraction and phonetic posteriorgrams for effective voice conversion. FreeVC, gaining attention for its robust performance, represents the latest innovation, showcasing improved conversion quality with minimal target speaker data.

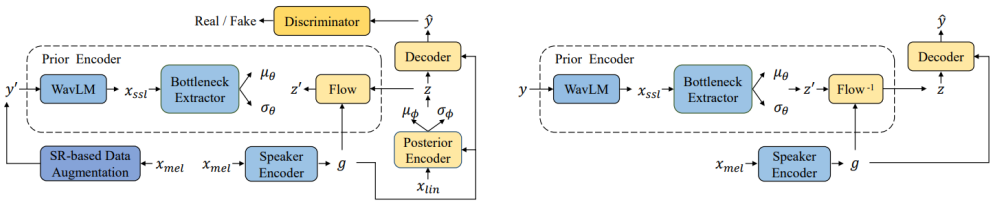


Figure 2.11: Training and inference procedure of FreeVC

Instead of using a two-stage reconstruction pipeline, FreeVC adopts the end-to-end framework of VITS for its high-quality reconstruction ability. It learns to to disentangle content information without the need of text annotation. Moreover, utilizes advancements in speech self-supervised learning (SSL), which has shown superior capabilities over traditional acoustic features like mel-spectrograms in various speech processing tasks. Specifically, FreeVC employs WavLM to extract SSL features directly from waveforms. Additionally, a bottleneck extractor is used to distill content information from these SSL features. To enhance the model's capacity to separate speaker identity from content, the paper introduces a novel data augmentation technique called spectrogram-resize (SR), which alters speaker characteristics without affecting the content. For one-shot voice conversion, FreeVC incorporates a speaker encoder that extracts speaker-specific information from a single audio sample.

While FreeVC achieves a commendable mean opinion score (MOS), indicating good naturalness, it doesn't perform as well in terms of speaker similarity as [5]. It is because FreeVC uses an information bottleneck and data augmentation to separate speaker identity from content. However, learning disentangled representations remains challenging. That makes the model struggle with achieving high speaker similarity, which is crucial for effective voice conversion.

2.5.2 Lightweight One-shot Voice Conversion

While previous any-to-any methods improve naturalness and speaker similarity, but at the cost of increased complexity As a result, training and inference have become more costly, making improvements difficult to evaluate and build on voice conversion models. To reduce the burden of training and inference procedure,

one solution we can think that is to simplify its architecture. Inspired by that idea, the model such as KNN-VC has taken a traditional approach such as unit selection and achieved promising results. First, these models use the aforementioned WavLM model to extract features from the input audio at the frame level, with each frame being approximately 20ms long. These frames have been shown in [18] to contain linguistic information. This information extraction is applied to both the source and target audio, and then an algorithm such as k-nearest neighbors (KNN) is used to find suitable ”candidates” from the target frames that can replace the source frames. Finally, these models synthesize the output audio using the previously selected frames. The biggest advantage of these models is that the output audio will have high similarity to the target audio because it is synthesized from the frames of the target speech. However, there will be drawbacks in terms of pronunciation with high Word Error Rate (WER).

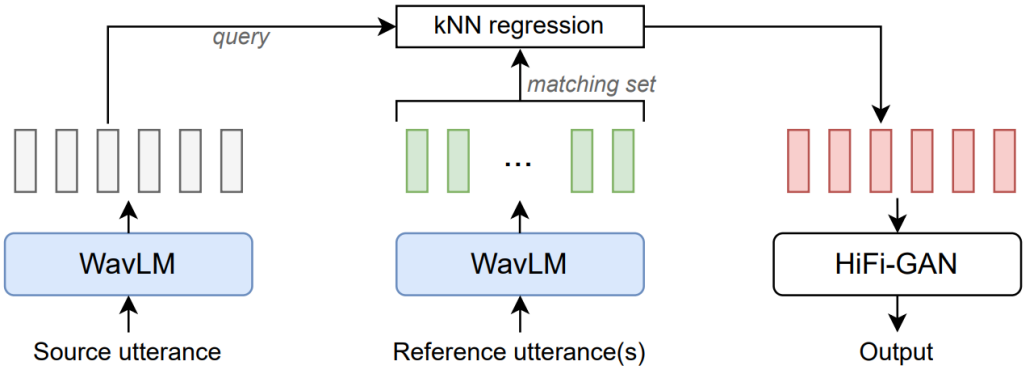


Figure 2.12: KNN VC - a voice conversion model with just nearest neighbors

In contrast with FreeVC, the kNN-VC model achieves better speaker similarity due to its direct construction from target speech, but performance in terms of intelligibility is compromised in scenarios where the target voice sample is limited. The primary reason for the reduced intelligibility with short target voice samples is attributed to the lack of sufficient target speech representations. When the target voice is short, only a limited number of self-supervised representations can be extracted, leading to a scenario where essential phonemes necessary for high-quality neighbor replacement during conversion might be missing.

The Phoneme Hallucinator model was developed to overcome the shortcomings of the KNN-VC model by introducing a conditional generative model that can hallucinate target representation features conditioned on a small target representation set. With more target features, the target neighbor candidates set is effectively expanded and the intelligibility of the converted speech is improved without

sacrificing the speaker similarity. For English, extensive experiments demonstrate that Phoneme Hallucinator achieves SOTA performance regarding both intelligibility and speaker similarity on one-shot VC tasks where only a 3-second of target speaker speech is given.

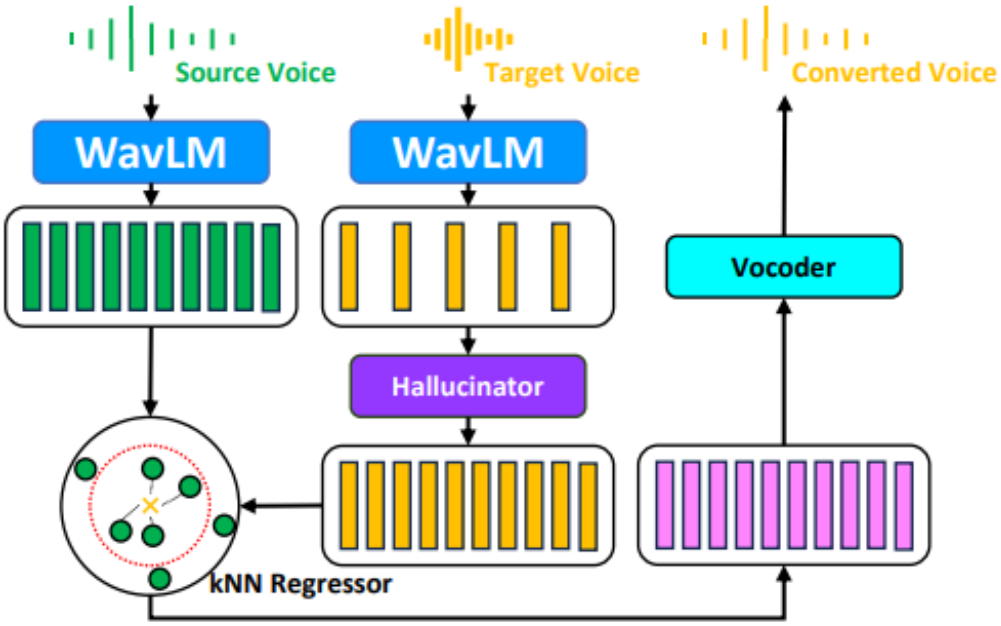


Figure 2.13: Phoneme Hallucinator: One-shot Voice Conversion via Set Expansion

With our experiments, the quality of these models for Vietnamese is not yet as good as expected because of two main reasons as mentioned in Chapter 1. That leads to deteriorates the quality severely about similarity. In the following chapters, we will discuss the difficulties of training voice conversion models for Vietnamese and the improvements that have been made.

CHAPTER 3. PROPOSAL OF VIETNAMESE VOICE CONVERSION MODEL

In Chapter 2, we dissected the spectrum of methodologies in voice conversion, spanning from the traditional signal processing approaches to the cutting-edge deep learning paradigms. We examined the advantages and limitations of traditional methods, such as codebook-based conversion and Gaussian Mixture Models (GMMs), and contrasted these with the advanced capabilities of deep learning techniques, including autoencoders, Generative Adversarial Networks (GANs), and sequence-to-sequence models. After that, we also presented the one-shot SOTA models such as KNN-VC [16], Phoneme Hallucinator [4]. This exploration underscored the impact of deep learning on the VC field, setting the stage for proposing a model specifically attuned to the nuances of the Vietnamese language.

3.1 Baseline Model

As discussed in previous chapters, the development of high-quality deep learning techniques has led to the continuous improvement and simplification of the data preparation and model training processes for voice conversion models. Therefore, choosing deep learning models for development in Vietnamese is a reasonable choice. Additionally, a practical voice conversion model requires minimal data, sometimes only a single sample, also known as one-shot, and exhibits high similarity to the target speaker. Based on these criteria and the foundation presented in Chapter 1 & 2 we decide to choose Phoneme Hallucinator [4] as the baseline model to develop a Vietnamese voice conversion model.

3.1.1 Model Architecture

As mentioned in Chapter 2, the Phoneme Hallucinator inherited the architecture from KNN-VC [16] which extracts frame-wise self-supervised representations for the source and the target speech. As the representations contain both speaker characteristics and phoneme information, conversion can be performed by directly replacing each frame of the source representation with its nearest neighbors in the target representation set. Finally, the audio is synthesized by a vocoder that takes the converted representation as input. Compared to speaker embedding VC methods, kNN-VC achieves better speaker similarity because the output is con-

structed exclusively from target speech. However, the intelligibility of kNN-VC deteriorates severely when the target voice is short as shown by its high WER [4]. This problem may negatively impact its real-world application where users usually can only provide several seconds of target speech. The cause of this problem is straightforward: for short target voices, only a small number of self-supervised representations can be extracted. As a result, some key neighbors with necessary phonemes are missing for high-quality neighbor replacement.

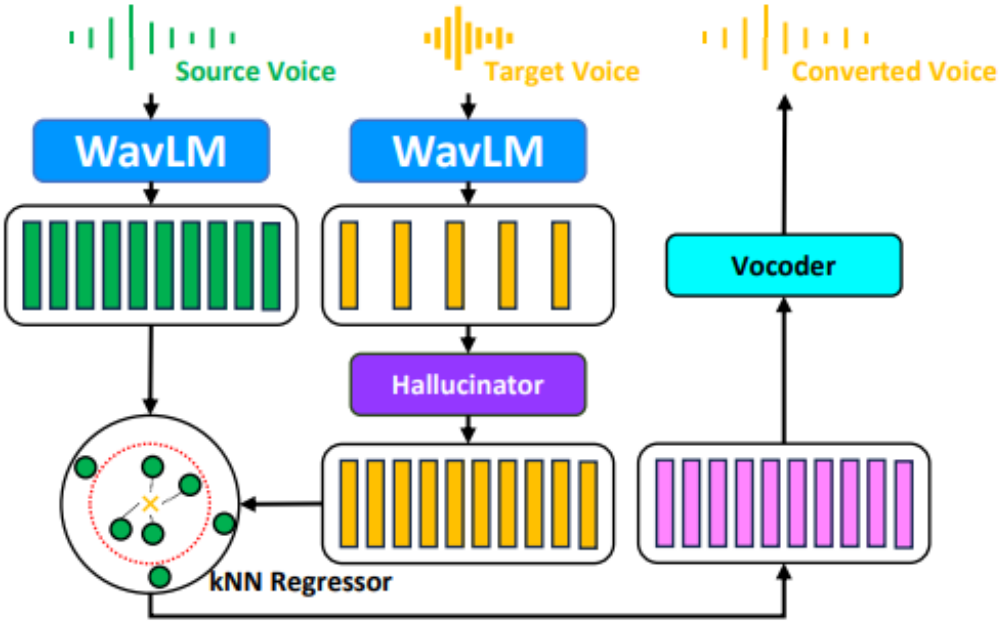


Figure 3.1: Phoneme Hallucinator Voice Conversion Pipeline [4]

In Phoneme Hallucinator, a conditional generative model that can hallucinate target representation features conditioned on a small target representation set was proposed. With more target features, the target neighbor candidates set is effectively expanded and the intelligibility of the converted speech is improved without sacrificing the speaker similarity. The VC pipeline presented in Figure 3.1 includes steps:

- (1) A pre-trained WavLM model [18] extracts the source representation sequence (green) or SSL features from the source voice and the target representation set (yellow) from the target voice respectively.
- (2) The target set is expanded by the proposed hallucinator.
- (3) Every source representation is replaced by its neighbors in the expanded target set, resulting in the converted sequence (pink).

- (4) Finally, a pre-trained vocoder transforms the converted sequence to voice.

The baseline model use SSL features which contain both speaker characteristics and phoneme information. And self-supervised learning (SSL) is a paradigm in machine learning where a model is trained on a task using the data itself to generate supervisory signals, rather than relying on external labels provided by humans. In the context of neural networks, self-supervised learning aims to leverage inherent structures or relationships within the input data to create meaningful training signals. SSL tasks are designed so that solving it requires capturing essential features or relationships in the data. The input data is typically augmented or transformed in a way that creates pairs of related samples. One sample serves as the input, and the other is used to formulate the supervisory signal. This augmentation can involve introducing noise, cropping, rotation, or other transformations. Self-supervised learning more closely imitates the way humans learn to classify objects.

3.1.2 Model Components

There are four main components in the Phoneme Hallucinator model:

- **WavLM** - a pre-trained representation learning model that was trained on hundreds of thousands of hours non-tonal data such as English. Which was used to extract SSL features from source & target audio.
- **Hallucinator** - a conditional generative model [34] that can hallucinate target representation features conditioned on a small target representation set.
- **KNN Regressor** - To find top k-nearest neighbors from the target set by KNN algorithm [35]
- **Vocoder** - To vocode the converted features with a neural vocoder to obtain the converted speech.

a, WavLM

WAVLM, which stands for Weighted Audio-Visual Language Model, is an advanced multi-modal model developed to enhance the integration and interpretation of audio and visual data within natural language processing tasks. This model leverages the strengths of audio-visual data to provide superior performance in tasks such as speech recognition, audio-visual speech recognition, and

cross-modal audio-visual understanding. The WAVLM [18] model is built upon a transformer-based architecture that efficiently processes and learns from both audio and visual inputs. The key components of the WAVLM architecture as shown in the Figure 3.2 include:

- **Audio Encoder:** Processes the audio signals to extract relevant features and representations.
- **Visual Encoder:** Analyzes visual data to capture important visual features that complement the audio data.
- **Fusion Layer:** Integrates the encoded audio and visual features to create a unified representation that is used for further processing and task-specific applications.

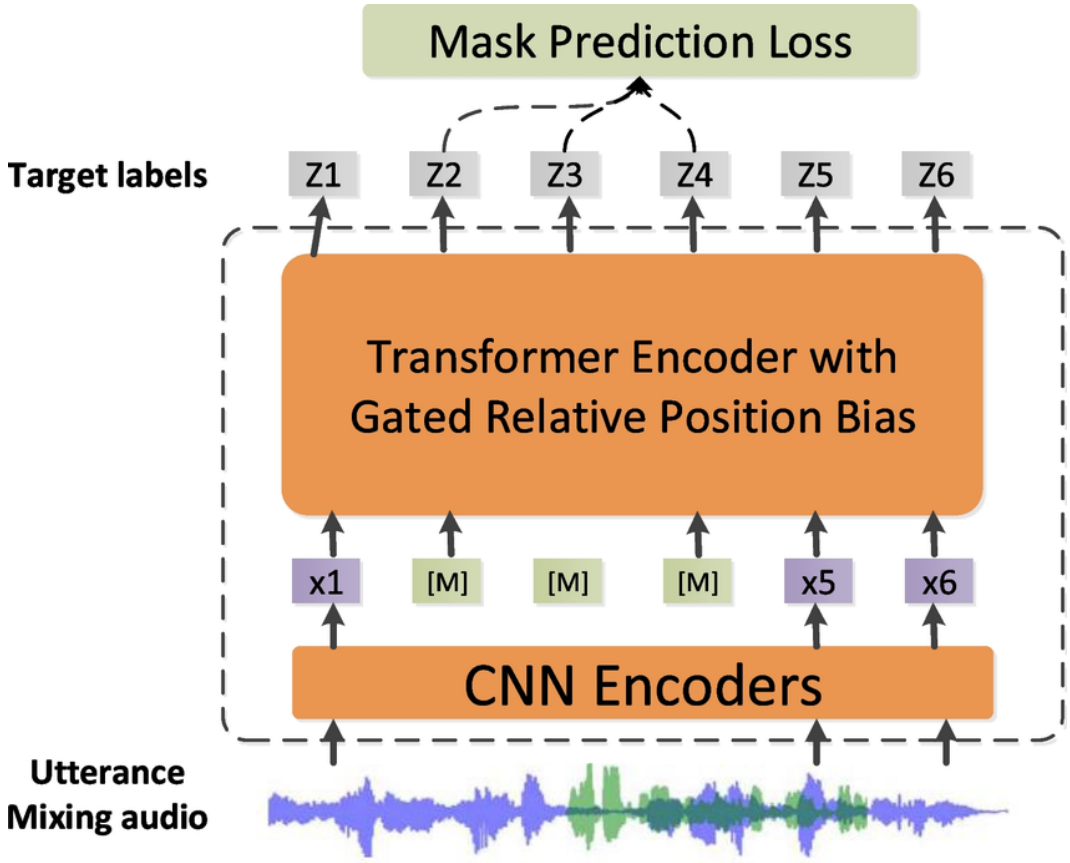


Figure 3.2: WavLM Architecture [18]

WavLM can be used to extract robust speaker-specific features from audio data. This includes capturing the nuances in speech that are unique to an individual's vocal tract and style of speaking. By fine-tuning WavLM on a dataset of spoken

audio, one can derive detailed embeddings that represent the acoustic properties of speech crucial for voice conversion. These features can then be fed into a voice conversion model, such as a variational autoencoder (VAE) [14] or a generative adversarial network (GAN) [15]. These models learn to map the extracted features from the source speaker to match the vocal characteristics of the target speaker. After the initial conversion, the output might need fine-tuning to eliminate any artifacts and ensure the speech sounds natural. Additional processing like pitch correction and formant adjustment might be required, depending on the application.

b, Hallucinator & KNN Regressor

The Phoneme Hallucinator is play a role as a key component of a voice conversion system that addresses intelligibility deterioration in one-shot VC when only a short sample of target speech is available. It is a conditional generative model that hallucinates additional speech representation features to enhance the conversion process. The Phoneme Hallucinator expands the set of target speech representations by generating new features that maintain the characteristics of the target speaker. This expansion helps in maintaining high intelligibility and speaker similarity in the converted speech.

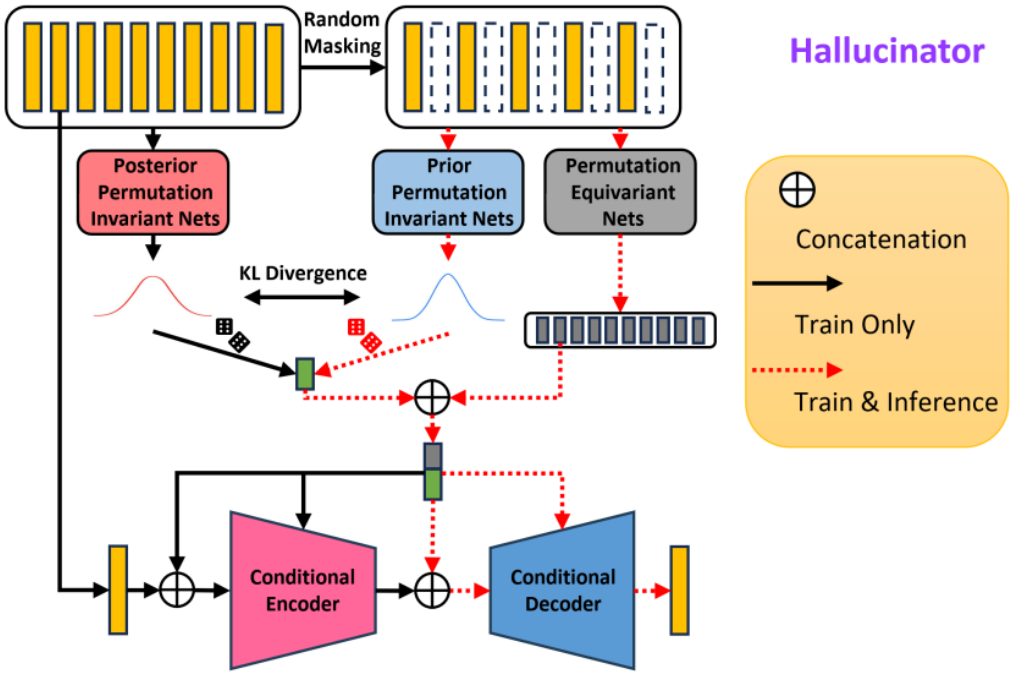


Figure 3.3: The detailed structure of the hallucinator [4]

The core functionality of the Phoneme Hallucinator is based on modeling the conditional distribution of the hallucinated features given a small set of actual target features. The conditional likelihood is given by:

$$p(X_e|X_t) = \int \left(\prod_{i=1}^{N_e} p(x_{e_i}|\theta) \right) p(\theta|X_t)\, d\theta, \tag{3.1}$$

where θ is a latent variable encapsulating speaker characteristics, X_t is the set of target features, and X_e is the set of hallucinated features. To make this feasible, the paper optimizes a variational lower bound of the log likelihood:

$$\log p(X_e|X_t) \geq \sum_{i=1}^{N_e} \mathbb{E}_{q(\theta|X_e,X_t)} \left[\log p(x_{e_i}|\theta)\right] - D_{KL}(q(\theta|X_e, X_t)\|p(\theta|X_t)), \quad (3.2)$$

where $q(\theta|X_e, X_t)$ is the approximate posterior distribution, and D_{KL} represents the Kullback-Leibler divergence [31]. The model employs permutation invariant and equivariant neural networks to respect the unordered nature of sets. A Set Transformer model is utilized to estimate the distributions involved effectively.

To train Hallucinator, we need to obtain an extra set of speech representations X_e conditioned on X_t. Equivalently, we can reformulate this task as obtaining a missing subset of speech representations X_e based on an observed subset of speech representations X_t. Both X_e and X_t are from the same speech utterance representation X with a fixed cardinality N where $X_e \cap X_t = \emptyset$ and $X = X_e \cup X_t$.

Though speech representations extracted from a continuous utterance are usually temporally correlated, we assume that the missing elements are Missing Completely at Random (MCAR). Our assumption is valid because our task is not to synthesize a sequence of temporally coherent speech representations, but to synthesize a set of unordered representations with diversified phonemes from the same speaker. Therefore, we obtain X_e and X_t during training as follows:

1. uniformly choose the number of missing representations N_e with $1 \leq N_e \leq N$.

2. uniformly select N_e elements from X, resulting in X_e.

3. mask out the elements selected above in X by zero vectors, resulting in X_t. This step is visualized in the top part of Fig. 3. To differentiate selected and un-selected elements, each element in X_t is concatenated with a binary mask.

After obtaining X_t and X_e, we can train our model by maximizing (2). During inference, our model takes X_t and outputs X_e. Note that we can run as many sampling steps as we want to obtain an infinite number of new samples.

The kNN algorithm is employed as part of a novel voice conversion pipeline.The kNN algorithm is applied as follows in the VC process: (1) Extract frame-wise speech representations for both the source and target speakers, (2) For each frame of the source speech, find the k-nearest speech representation frames from the

target speaker based on a defined similarity metric and (3) Replace the source speech representation with the average of its k-nearest neighbors from the target, thus preserving the linguistic content while adapting the speaker characteristics.

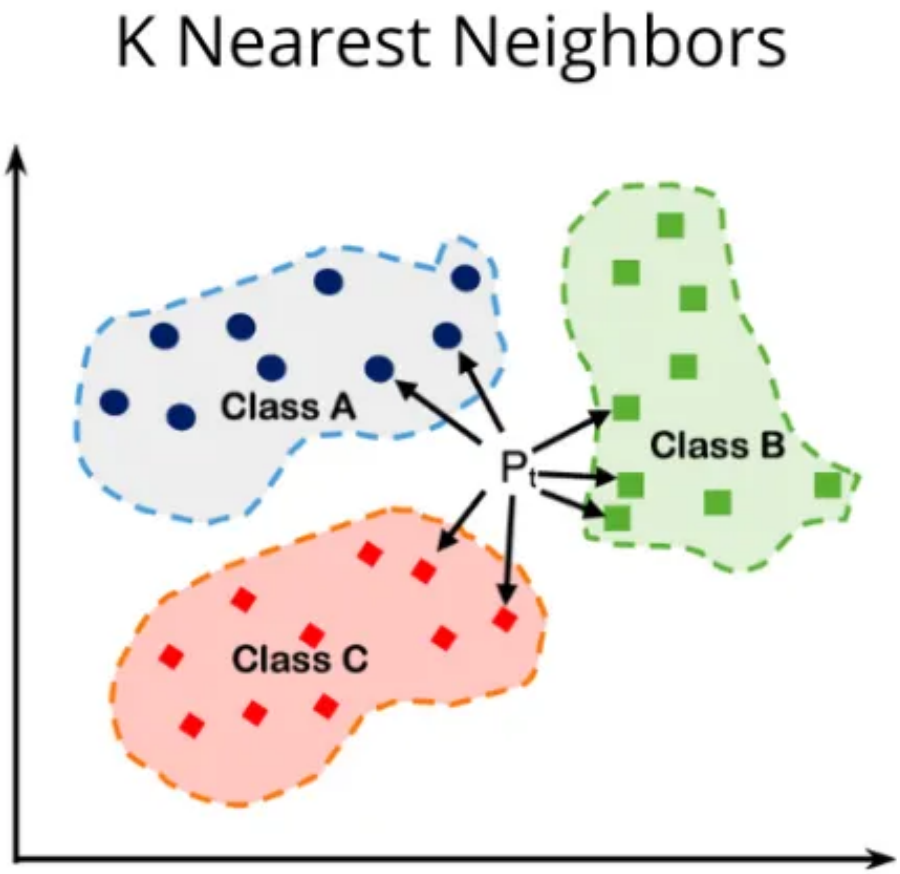


Figure 3.4: KNN Algorithm [36]

While effective, the basic kNN VC method faces challenges particularly when the target voice sample is very short, limiting the availability of diverse speech representations. This limitation is addressed in the paper by introducing a ”Phoneme Hallucinator” component, which expands the set of target speech representations

c, Vocoder

The HifiGAN architecture, with its impressive capabilities in generating high quality audio signals, can be modified to serve as a vocoder for voice conversion. The HifiGAN architecture consists of a generator and two discriminator networks The generator produces synthetic speech signals similar to original speech signals. It takes a low-dimensional noise vector as input and outputs a high-dimensional 39 speech signal. The generator uses convolutional layers to extract

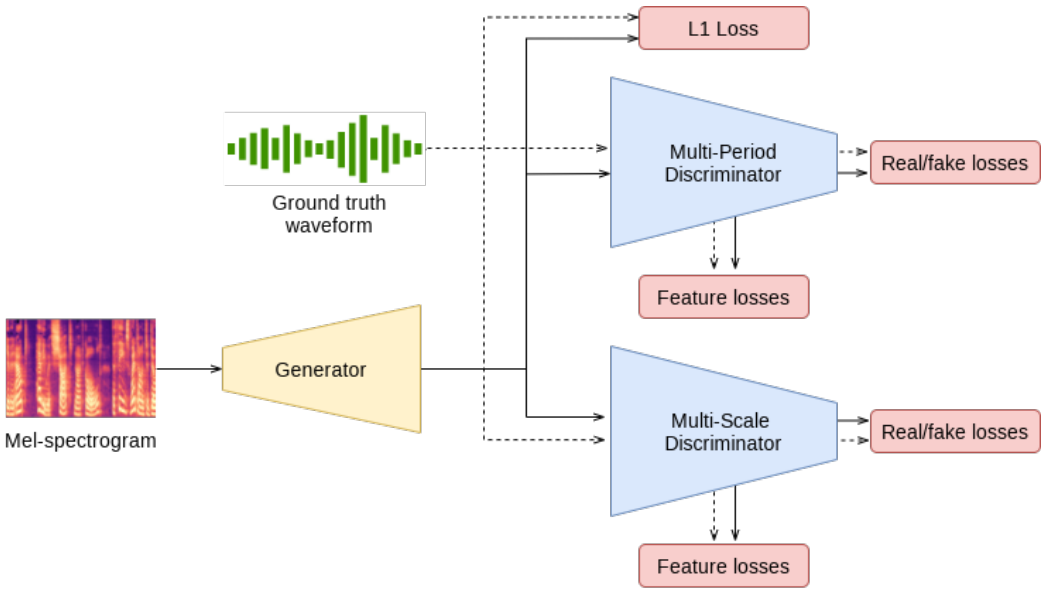


Figure 3.5: The detailed structure of the Hifi-GAN vocoder [37]

features from the input noise vector and transform them into a speech signal. The discriminators distinguish between real and synthetic speech signals. It takes real and synthetic speech signals as input and produces a binary classification output. The Multi-Period Discriminator is a sub-network that extracts patterns from the data that occur periodically (for example, every N time steps). The Multi-Scale Discriminator is a sub-network that extracts patterns from the data at different resolutions of the input signal. Lower resolutions are obtained by average-pooling the signal. Both discriminators can be further divided into smaller sub-networks that work at different resolutions. The loss functions take as inputs intermediate feature maps and the outputs of those sub-networks. The discriminators use convolutional layers to extract features from input speech signals and transform them into a binary classification output. The networks train simultaneously in an adversarial manner, with the generator generating realistic audio signals to fool the discriminators and the discriminators distinguishing real from fake audio signals. After training, the generator is used for synthesis, and the discriminators are discarded. All three components are convolutional networks with different architectures. One key advantage of the HifiGAN architecture is its ability to generate highquality speech signals with minimal distortion. This uses a multi-scale feature extraction scheme that captures both short-term and long-term dependencies in the input signal. Another advantage is efficiently learning from large datasets using a hierarchical training procedure, which allows the model to gradually learn more complex features through network layers.

3.2 Challenges for Vietnamese Voice Conversion

3.2.1 Tonal Language

Vietnamese is a tonal language spoken by approximately 90 million people primarily in Vietnam, as well as in diaspora communities around the world. It belongs to the Austroasiatic language family, specifically the Vietic branch, and is heavily influenced by Chinese, due to centuries of cultural and linguistic interaction. Tonal languages are characterized by the use of pitch variations, or tones, to distinguish word meanings. In Vietnamese, there are six distinct tones: flat, rising, falling, low-dipping, high-dipping, and low-rising. The tones can drastically change the meaning of a word, making correct pronunciation essential for effective communication. The Vietnamese writing system is based on the Latin alphabet with additional diacritics to indicate tones. This orthographic system, known as Quoc Ngu, was introduced by French missionaries in the 17th century and has since become the official writing system of Vietnam. [38]

Studying the Vietnamese language, especially its phonetics and phonology, is necessary to understand a language as a means of communication between people; hence plays important roles in speech processing. This section recapitulates the phonetic and phonology of the modern Hanoi, which is widely considered as the standard language of Vietnamese.

Vietnamese syllables are elementary units that have one way of pronunciation. In documents, they are usually delimited by white-space. Being the elementary units, Vietnamese syllables are not undivided elements but a structure [39]. Figure 3.6 depicts the general structure of Vietnamese syllable

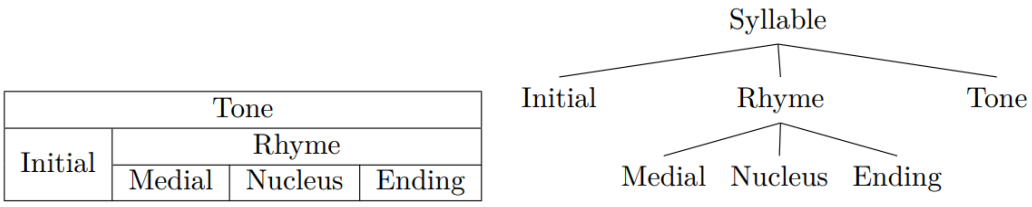


Figure 3.6: Vietnamese Syllable Structure [39]

Generally, each Vietnamese syllable has two main parts: an initial consonant and a rhyme. A tone appears simultaneously with three segmental elements of rhyme, i.e. medial, nucleus and ending. The nucleus and tone are compulsory while others are optional [39]

- **Initial (Onset):** This is the beginning consonant or consonant cluster of the syllable. In Vietnamese phonology, the initial consonant plays a significant role in the pronunciation and meaning of the syllable.
- **Rhyme:** The rhyme can be broken down into three parts:
  - **Medial:** A glide that occurs after the initial and before the nucleus. Not all syllables have a medial.
  - **Nucleus:** Typically a vowel or diphthong that is the core part of the syllable and carries the main vocalic sound.
  - **Ending:** This can be a final consonant or a phonetic feature that occurs at the end of the syllable.
- **Tone:** Vietnamese is a tonal language, and the tone is an essential feature that overlays the entire syllable. It dictates the pitch contour and can change the meaning of the word.

From that we can see Vietnamese has linguistic diversity and tonal language specificity. So, pre-trained models in other languages cannot easily grasp this diversity, making it difficult to achieve a one-size-fits-all solution for VC in Vietnamese. Moreover, with specific intonation patterns that are pivotal for meaning. The lack of pre-trained models that understand and replicate these tonal intricacies hinders the creation of VC systems capable of accurately transforming speech without losing semantic content.

3.2.2 Data Limitations

The training of robust pre-trained models requires vast amounts of data. For Vietnamese, such datasets are not as readily available as for languages like English or Mandarin, which limits the ability to develop models that are well-generalized and capable of high performance. As mention in Chapter 1 most of the existing research in Vietnam has been primarily focused on basic speech processing tasks, such as speech recognition and synthesis, with relatively fewer studies dedicated to voice conversion. Several speech processing challenges have been organized such as Zalo AI Challenge [19], VLSP [20] but there are no shared tasks for voice conversion. Furthermore, the lack of publicly available Vietnamese speech corpora and benchmark datasets such as VTCK [2], LibriSpeech [3] which poses a significant challenge for researchers in Vietnam, hindering the development and evaluation of voice conversion systems tailored to the Vietnamese language.

3.3 Proposed Vietnamese Voice Conversion Model

From the difficulties previously mentioned as well as analyses specific to Vietnamese, we have put forward the following improvement proposals:

- (1) Propose a new Text2SSL module based on FastSpeech2 [40] acoustic model to get more linguistic information from the source text. This component is crucial for adding tonal features that were not presented with the features extracted by WavLM. These features will be combined with the WavLM feature to improve tonal problems in the output speech.
- (2) Apply augmentation technique idea from FreeVC [5] to create a diverse multi-speakers dataset. This helps the model more robust when performing one-shot voice conversion, because the vast of data.
- (3) Instead of using the pre-trained vocoder which was only trained with English datasets. We trained this vocoder on the dataset created in step (2) that only contains Vietnamese audio.

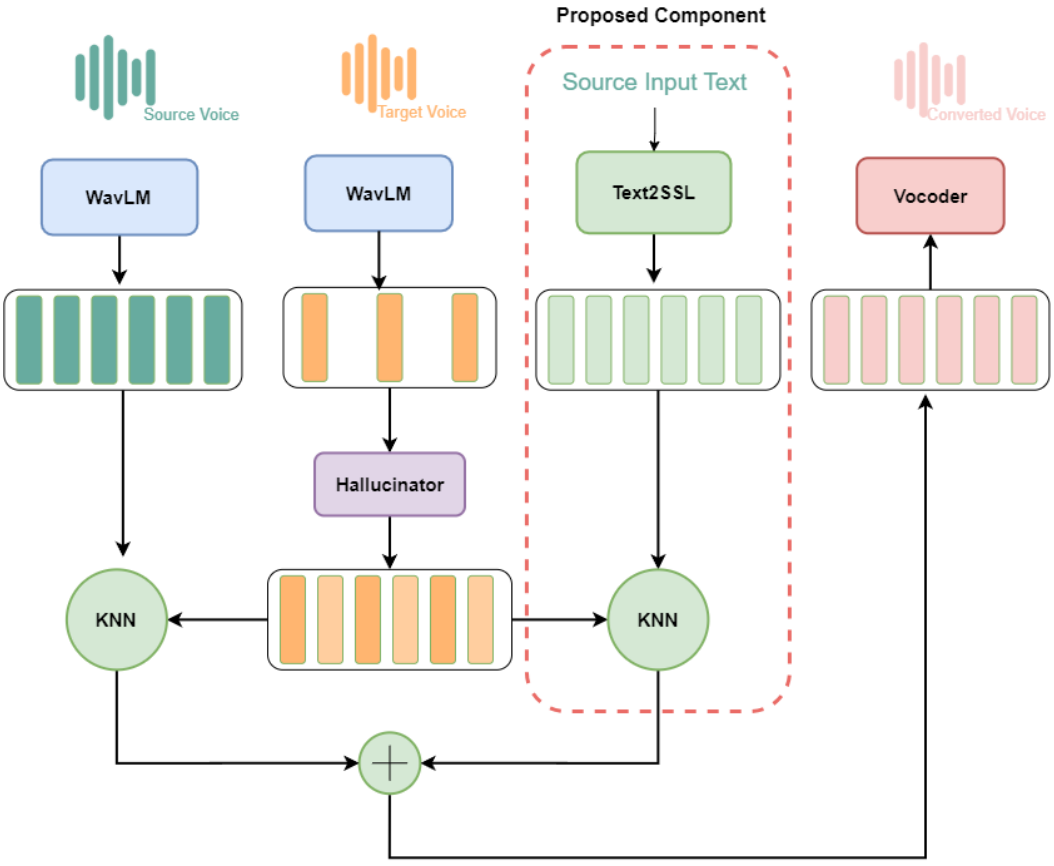


Figure 3.7: Proposal of Vietnamese Voice Conversion Model

3.3.1 Text2SSL Component

a, FastSpeech 2

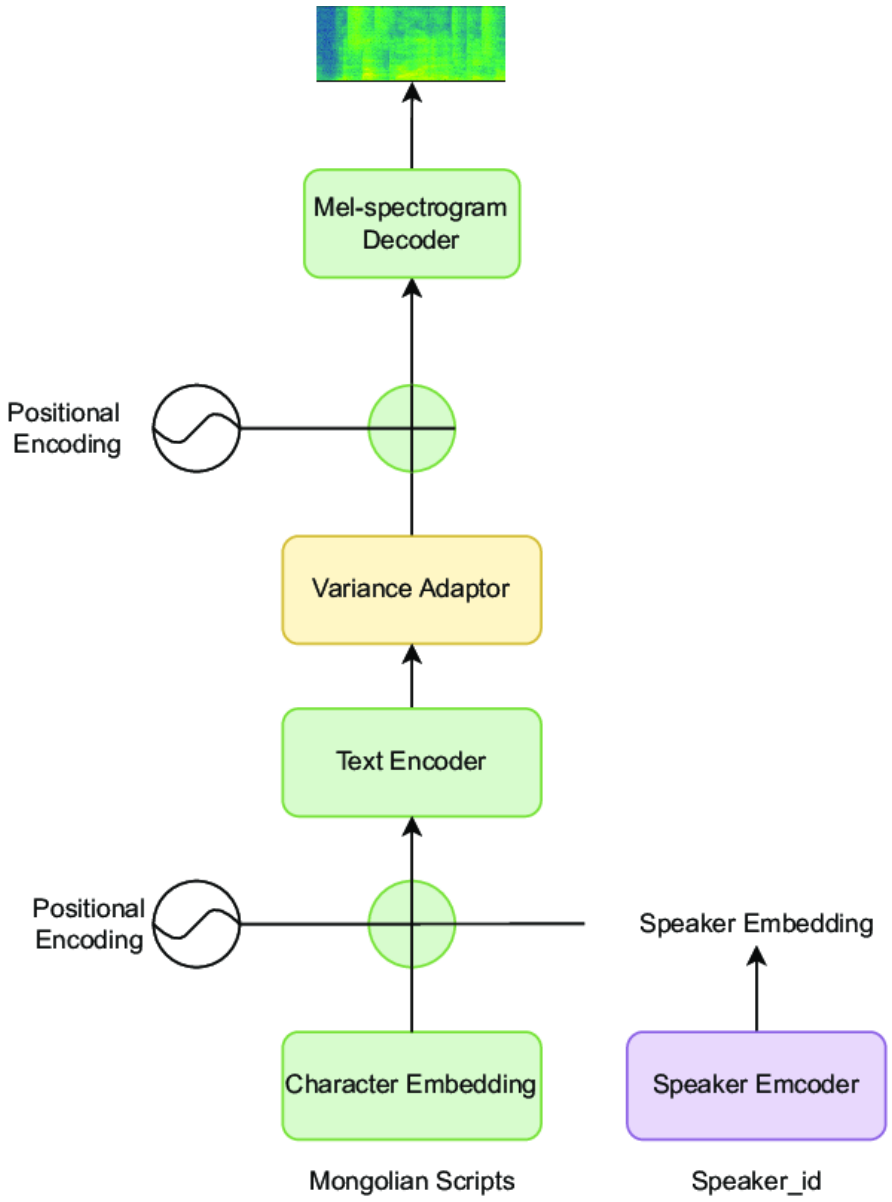


Figure 3.8: FastSpeech2 Architecture [40]

FastSpeech2 [40] is an advanced text-to-speech (TTS) model that enhances its predecessor, FastSpeech, by incorporating variance adaptors to improve control over prosody, thus allowing adjustments in pitch, duration, and energy for more natural-sounding speech. It maintains the robustness and speed of the original model, generating speech faster than real-time, which makes it ideal for applications requiring immediate speech output such as virtual assistants and audiobook narration. The key improvements include clearer speech production, explicit pre-

dictions of phoneme duration, pitch, and energy, and the use of effective training techniques like self-supervised learning. These features make FastSpeech 2 a significant advancement in TTS technology, offering both speed and flexibility in speech generation while avoiding common issues like word skipping and repetition seen in traditional autoregressive models [41].

In voice conversion, instead of using text & mel-spectrogram [24] as the training inputs, we replace mel-spectrogram with self-supervised-learning features. That means when we put the text into the model we will get the SSL features [18]. Unlike the SSL features extracted from WavLM, which only contain information about prosody, these features focus on linguistic characteristics, specifically Vietnamese. Therefore, they will include tonal information tailored specifically for Vietnamese. This significantly reduces the loss of tonal phenomena of the original model.

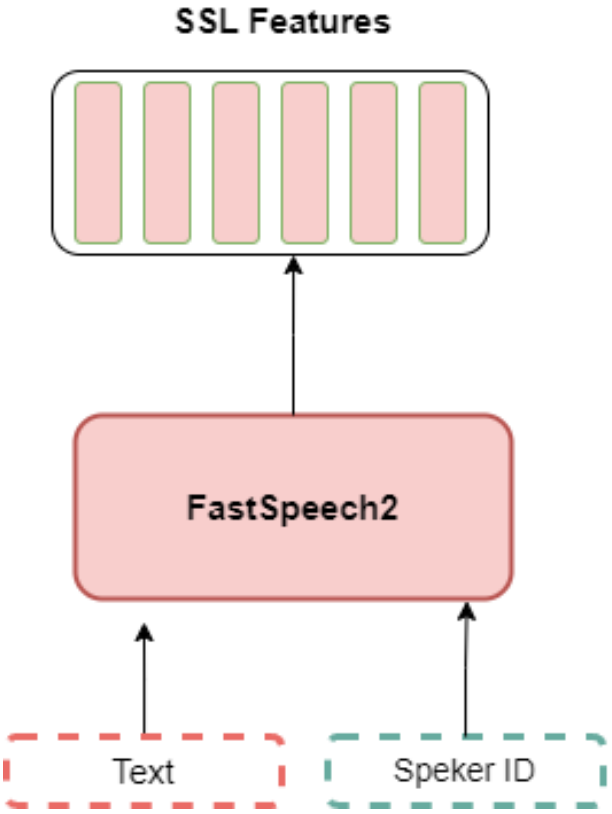


Figure 3.9: The Architecture of Text2SSL Component

With the features generated from the source voice through WavLM and the features generated by the Text2SSL module from the source text input, there will be a mismatch in the size of the two vectors due to the different dimensions between text and audio. To combine them, we need to align these vectors so that

they have the same length. The algorithm used here involves randomly repeating adjacent frames. Experiments show that this method does not significantly affect the model's output quality

3.3.2 Create A Diverse Multi-Speakers Dataset

In FreeVC [5], the Spectrogram-Resize (SR) based augmentation, designed to improve the disentanglement ability of a voice conversion model, FreeVC. This technique manipulates the speaker information in source waveforms without altering the content information, thereby enhancing the model's ability to cleanly separate content from speaker characteristics. Specifically, the augmentation process involves obtaining a mel-spectrogram from a waveform, conducting a vertical SR operation to adjust the mel-spectrogram vertically by a certain ratio using bilinear interpolation, and then either padding or cutting it to match the original shape. This approach allows the model to learn more effectively from augmented speech, retaining content information while varying the speaker-specific features like pitch and formant distance.

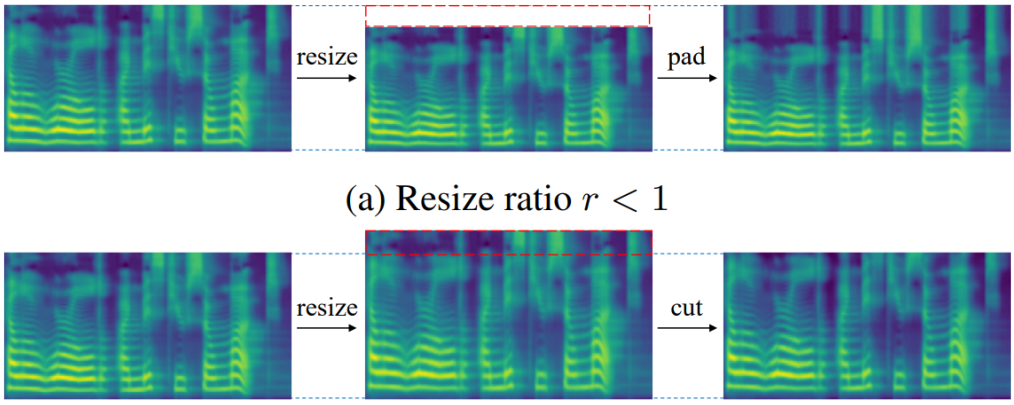


Figure 3.10: The 2 Spectrogram Augmentation Techiques: (1) Cut the top frequency, (2) Pad the top frequency [5]

A mel-spectrogram can be seen as an image with horizontal time axis and vertical frequency bin axis. Vertical SR operation first resizes mel-spectrogram to a certain ratio r vertically using bilinear interpolation, and then pads or cuts the resized mel-spectrogram to the original shape. If the ratio r is less than 1, It pad the squeezed mel-spectrogram at the top with the sum of highest frequency bin value and Gaussian noise, resulting in speech with lower pitch and closer formant distance; else, we cut redundant frequency bins at the top of the stretched

mel-spectrogram, resulting in speech with higher pitch and farther formant distance. By training with augmented speech, the model will better learn to extract the unchanged content information shared in each ratio r.

In this thesis, we utilize the augmentation technique described above to create a multi-speaker dataset. Same with FreeVC [5], for each audio sample, we apply resize augmentation 5 times, and for each augmentation, we have 6 frequency levels for cutting or padding. This means that for each original speaker, we can generate 30 speaker variances. With the original dataset for Vietnamese, we have a total of 15 speakers. After completing the augmentation process, we have a multi-speaker dataset with 450 speakers. Moreover, we will this dataset to fine-tune the Hifi-GAN [37] vocoder for the Vietnamese language. Details regarding the dataset information before and after augmentation are described in two tables below:

Table 3.1: Dataset Summary before applying SR Augmentation

Total Audio	6957
Total Speaker	15
Audio per Speaker	464
Mean Duration (s)	7.34
Min Duration (s)	0.19
Max Duration (s)	58.75

Table 3.2: Dataset Summary after applying SR Augmentation

Total Audio	208710
Total Speaker	450
Audio per Speaker	464
Mean Duration (s)	7.34
Min Duration (s)	0.19
Max Duration (s)	58.75

By incorporating data from multiple speakers, firstly the dataset enhances the model's ability to generalize to unseen speakers and capture the variability present in natural speech, thus improving its generalization. Secondly, training on a diverse set of speakers makes the model more robust to variations in speaker characteristics such as pitch, timbre, and speaking style, ensuring reliable performance across a wide range of speakers not present in the training data. Lastly, with a larger and more diverse dataset, voice conversion models can explore the speaker space more comprehensively, potentially uncovering new insights and patterns in the speech representations. The quality of the audio after using the augmentation

method will not be affected because this method only changes the top frequency of the audio signal without adding noise or distorting the original sound.

4.1 Evaluation Strategy

4.1.1 Evaluation Metrics

Effective quality assessment of voice quality is required to validate the algorithms, to measure the technological progress, and to benchmark a system against the state-of-the-art. Typically, we report the results in terms of objective and subjective measurements.

- To provide an objective evaluation, a reference speech is required. Objective evaluation, on the other hand, uses objective measures and replaces human test subjects with signal processing processes to algorithmically evaluate the synthetic signal, as discussed. The common objective evaluation metrics include Mel-cepstral distortion (MCD) [42] for spectrum, and RMSE [43] for prosody. Such metrics are not always correlated with human perception partly because they measure the distortion of acoustic features rather than the waveform that humans actually listen to.
- Subjective evaluation metrics, such as the mean opinion score (MOS) which collects opinions from human test subjects in a standardized procedure (as outlined in [67]) could represent the intrinsic naturalness and similarity to the target. For subjective evaluation to be meaningful, a sufficient number of listeners are required.

Because the two primary criteria for evaluating a voice conversion model are content intelligibility and speaker similarity, and with the aim of simplifying this thesis, we will use three metrics, namely MOS, WER, and SIM, to assess the quality of the final model.

- Mean Opinion Score (MOS) [44]: The naturalness of a voice conversion refers to how closely it resembles human speech. A high-quality voice conversion system should sound like a real human being, with appropriate pauses, intonation, and emphasis. The Mean Opinion Score (MOS) was used for this aspect. Listeners evaluated emotional synthetic speech using a 5-point scale:
  - 1. Bad (Very annoying and objectionable)

  - 2. Poor (Annoying, but not objectionable)
  - 3. Acceptable (Perceptible and slightly annoying)
  - 4. Good (Just perceptible, but not annoying)
  - 5. Excellent (Imperceptible).

- Word Error Rate (WER) [45] serves as a crucial metric for assessing the accuracy of converted speech in preserving linguistic content. WER quantifies the disparity between the converted speech and a reference transcript by measuring the number of substitutions, deletions, and insertions required to align them. A lower WER indicates higher accuracy, reflecting a closer resemblance between the converted speech and the reference. By incorporating WER into the evaluation process, researchers can objectively measure the system's ability to maintain content intelligibility and linguistic fidelity during conversion.

- Similarity (SIM) [46]: assesses how closely the converted speech resembles the original speaker's voice. SIM is calculated by comparing acoustic features extracted from the original and converted speech, such as pitch and spectral envelope. Methods include cosine similarity, Euclidean distance, or machine learning models trained to predict similarity scores. A higher SIM score indicates greater similarity, reflecting successful preservation of the speaker's identity during conversion. Incorporating SIM into evaluation provides insights into the fidelity of voice conversion beyond linguistic accuracy.

4.1.2 Evaluation Design

4.2 Experimental Setup

We trained four models including (1) Text2SSL, (2) KNN-VC, (3) Phoneme Hallucinator and (4) HiFi-GAN vocoder. The audio format is 16-bit PCM with a 22,050 Hz sampling rate. The FFT, window, and hop size were set to 1024, 1024, and 256, respectively, before being converted into mel-spectrograms following [24]

4.2.1 Model Configuration

a, Text2SSL's Configurations

To training Text2SSL module, we use almost the same config as FastSpeech2. The Conformer module with the same number of layers - 4 blocks in the Encoder and the Mel-spectrogram Decoder - replaces the Transformer module. The number of attention heads is set to 2. The size of the phoneme vocabulary is 45,176, including punctuation. The kernel sizes of the 1D-convolution are set to 3 in the Variance Predictor, with input/output sizes of 256/256 for both layers and a dropout rate of 0.1. To improve the overall reconstructed mel-spectrogram, the Post-Net layer described in the Tacotron2 paper is leveraged. The predicted mel-spectrogram is passed through a 5-layer convolutional Post-Net. Each Post-Net layer is composed of 256 filters with the shape 5×1 with batch normalization, followed by Tanh activations on all but the final layer.

Table 4.1: Text2SSL's Model Configuration

Parameter	**Value**
transformer.encoder_layer	4
transformer.encoder_head	2
transformer.encoder_hidden	256
transformer.decoder_layer	6
transformer.decoder_head	2
transformer.decoder_hidden	256
transformer.conv_filter_size	1024
transformer.conv_kernel_size	[9, 1]
transformer.encoder_dropout	0.2
transformer.decoder_dropout	0.2
variance_predictor.filter_size	256
variance_predictor.kernel_size	3
variance_predictor.dropout	0.5
variance_embedding.pitch_quantization	”linear”
variance_embedding.energy_quantization	”linear”
variance_embedding.n_bins	256
multi_speaker	False
max_seq_len	1000
optimizer.batch_size	16
optimizer.betas	[0.9, 0.98]
optimizer.eps	0.000000001
optimizer.weight_decay	0.0
optimizer.grad_clip_thresh	1.0
optimizer.grad_acc_step	1
optimizer.warm_up_step	4000
optimizer.anneal_steps	[300000, 400000, 500000]
optimizer.anneal_rate	0.3

b, KNN-VC's Configurations

To train KNN-VC, we used the configs for each module as the following:

- **Encoder**: We use the pretrained WavLM-Large encoder as mentioned in the paper from **6** to extract features for the source and reference utterances.
- **KNN regression**: We set $k = 4$ with a uniform weighting and use cosine distance to compare features. In preliminary experiments, we found that kNN-VC is quite robust to a range of values around $k = 4$. Namely, when more reference audio is available (e.g. ≥ 10 mins), the conversion quality may even be improved by using larger values of k (in the order of $k = 20$).

Parameter	Value
resblock	1
num_gpus	0
batch_size	16
learning_rate	0.0002
adam_b1	0.8
adam_b2	0.99
lr_decay	0.999
seed	1234
upsample_rates	[10, 8, 2, 2]
upsample_kernel_sizes	[20, 16, 4, 4]
upsample_initial_channel	512
resblock_kernel_sizes	[3, 7, 11]
resblock_dilation_sizes	[[1, 3, 5], [1, 3, 5], [1, 3, 5]]
hubert_dim	1024
hifi_dim	512
segment_size	7040
num_mels	80
num_freq	1025
n_fft	1024
hop_size	320
win_size	1024
sampling_rate	16000
fmin	0
fmax	8000
fmax_for_loss	null
num_workers	4

Table 4.2: KNN-VC's Configuration Parameters

c, Vocoder's Configurations

Instead of using the HiFi-GAN V1 which was trained on English only we used HiFi-GAN V2 which was pretrained on Vietnamese augmented datasets.

Table 4.3: Hifi-GAN Vocoder's Configuration Parameters

Parameter	**Value**
resblock	1
num_gpus	0
batch_size	16
learning_rate	0.0002
adam_b1	0.8
adam_b2	0.99
lr_decay	0.999
seed	1234
upsample_rates	[8,8,2,2]
upsample_kernel_sizes	[16,16,4,4]
upsample_initial_channel	128
resblock_kernel_sizes	[3,7,11]
resblock_dilation_sizes	[[1,3,5], [1,3,5], [1,3,5]]
segment_size	8192
num_mels	80
num_freq	1025
n_fft	1024
hop_size	256
win_size	1024
sampling_rate	22050
fmin	0
fmax	8000
fmax_for_loss	null
num_workers	4
dist_config.dist_backend	nccl
dist_config.dist_url	tcp://localhost:54321
dist_config.world_size	1

d, Phoneme Hallucinator's Configurations

To train Phoneme Hallucinator model, we only use utterances longer than 4 seconds, which ensures that we can extract a representation set with a cardinality larger than 200. We randomly select a subset of 200 represents to constitute X, which will be divided into the missing subset X_e and the observed subset X_t to train our model as discussed in Sec. 3.3. During every inference step, if the cardinality N_t of the target set is greater than 100, we uniformly select a subset with a cardinality of 100 to form X_t. To facilitate learning, we divide the representations by 10 to normalize them to have a standard deviation of around 1. All Set Transformers contain 4 multi-head attention blocks with a hidden size of 256 and 16 induction points. The sizes of θ and g_i are 256. The size of z is 256 and both

the decoder and the encoder in Eq. (3) is a 4-layer MLP with a hidden size of 512. We do not use any normalization layer for both MLP and Set Transformer.

Parameter	**Value**
dataset	speech
dimension	1024
batch_size	50
set_size	200
mask_type	arb_expand
model	pc_acset_vae
latent_encoder_hidden	[256,256,256,256]
latent_dim	256
trans_params.transform	[“L”,“LR”,“CP”]
trans_params.dimension	256
trans_params.coupling_hids	[256,256]
vae_params.hid_dimensions	256
vae_params.dimension	1024
vae_params.enc_dense_hids	[512,512,512,512]
vae_params.dec_dense_hids	[512,512,512,512]
use_peq_embed	1
set_xformer_hids	[256,256,256,256]
epochs	1000
optimizer	adam
lr	0.0001
decay_steps	100000
decay_rate	0.5
clip_gradient	1
summ_freq	100
eval_metrics	[“sam”]

Table 4.4: Phoneme Hallucinator’s Configuration Parameters

4.2.2 Training Settings

All models are trained using GPU 24GB geforce 4090 running on NVIDIA DGX 2 machines with automatic mixed precision. Acoustic models are trained up to 4,000 epochs with a batch size of 256 sentences. The Adam optimizer [47] is used with a learning rate of 10^{-3}, $\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-9}$. The learning rate is increased for a warm-up period of 3,000 steps and then decayed according to the Conformer schedule. The gradient accumulation trick is utilized to increase the batch size due to the limited VRAM of the GPU. Training a new vocoder for each speaker is impossible due to the limited speech data available in each dataset. To help the model reach convergence faster, a pre-trained HiFi-GAN model on the LJ speech dataset is leveraged as the starting point of the model. The HiFi-GAN vocoder is trained up to 1,000,000 iterations for each speaker using weight normalization and an Adam optimizer with a fixed learning rate of 2×10^{-4} with a batch size of 16 samples. The networks were trained using the AdamW optimizer [47] with $\beta_1 = 0.8$, $\beta_2 = 0.99$, and weight decay $\lambda = 0.0$. The Hallucinator [47] was implemented using TensorFlow by authors and we trained with the Adam optimizer. It employs a learning rate of 1×10^{-4} and a batch size of 50 over 250 epochs. Training takes approximately 6.5 hours. Our model maximizes the log-likelihood function as defined in equation (2), with a total of 19.03 million trainable parameters.

4.3 Result and Discussion

4.3.1 Evaluation Setup & Results

We conduct evaluations both subjectively and objectively for the proposed Vietnamese voice conversion model. For subjective evaluation, 15 participants are invited to evaluate the naturalness and speaker similarity of the speech in terms of 5-scale mean opinion score (MOS) and similarity mean opinion score (SMOS) respectively. For objective evaluation we use the WER metric with the text obtained with a public ASR model [48]. We also conduct experiments and evaluations with two other one-shot models: FreeVC [5] & KNN-VC [16] to prove the statements presented in the previous chapters. For all the mentioned evaluations we used the 10 reference audio which was recorded by smartphone by 10 different people who were not in the training dataset, to clean the environment noise in the recorded audio we used a public model called music-source-separation [49]. The details of evaluation results are presented as the following table:

Table 4.5: Performance Comparison of Voice Conversion Models (wo - stands for "without")

Model	WER (%)	MOS	SIM
kNN-VC (eng)	41.51	2.82 ± 0.17	3.50 ± 0.12
kNN-VC (vi)	50.92	2.55 ± 0.17	3.35 ± 0.12
FreeVC (eng)	6.52	4.05 ± 0.09	2.55 ± 0.11
FreeVC (vi)	7.45	3.95 ± 0.08	2.20 ± 0.11
Phoneme Hallucinator (baseline-eng)	5.68	4.05 ± 0.09	3.55 ± 0.12
Phoneme Hallucinator (baseline-vi)	6.24	2.75 ± 0.08	2.65 ± 0.12
Proposed Model (wo aug)	7.12	3.55 ± 0.08	3.15 ± 0.12
Proposed Model (with aug)	**5.35**	$\mathbf{3.90 \pm 0.08}$	$\mathbf{3.35 \pm 0.12}$

4.3.2 Discussion

From the results, we can discern that all original models, when trained for Vietnamese, yield inferior outcomes compared to models trained on English datasets. The reasons for this have been delineated in the preceding chapters, such as the WavLM model being trained predominantly on non-tonal data, or English datasets typically being large and encompassing thousands of speakers. This enables voice conversion models to be more robust and exhibit better generalization with unseen data.

We can observe that the kNN-VC model yields the poorest outcomes in terms of WER and MOS, a fact that has been explained in Chapter 2 due to the inadequacy of the matching set. However, it scores fairly well in SIM because it is synthesized solely from frames of the target audio, making it reasonable that the output voice resembles the target. Conversely, the FreeVC model demonstrates the opposite, with commendable MOS and WER scores but a lower SIM score, which aligns with analyses presented in previous chapters. As for the baseline model, it exhibits poor MOS performance, which has been clearly attributed to issues remaining for tonal languages such as Vietnamese. Its SIM score is marginally better than FreeVC but still falls short of kNN-VC.

For our proposed model, we conducted evaluations using both the version with the augmentation method and the one without. The detailed results are tabulated in Table 4.5. The evaluation results reveal that the proposed model, without the augmentation method, performs slightly worse in WER compared to the baseline model. However, it significantly outperforms the baseline in terms of SIM and MOS scores. This confirms our hypotheses and demonstrates that the proposed Text2SSL module has indeed improved the quality for tonal languages such as Vietnamese. Furthermore, the application of the augmentation method has enhanced the output quality, as specifically indicated by improvements across all three scores. This is understandable, as training on a diverse set of speakers makes the model more robust to variations in speaker characteristics such as pitch, timbre, and speaking style, ensuring reliable performance across a broad spectrum of speakers not represented in the training data.

In addition, through the experiments conducted in this thesis, we found that the audio quality is still distorted, which is due to the vocoder model. With insufficient and non-diverse training data, the output audio often contains noise such as metallic sounds or distortion. If we increase the amount of data, this issue will be improved significantly.

CHAPTER 5. CONCLUSIONS

This thesis outlined the development of a Vietnamese voice conversion model leveraging deep learning techniques to address the unique challenges posed by tonal languages and data limitations. The proposed model incorporates a Text2SSL module for enhanced linguistic context and employs a Spectrogram-Resize augmentation technique to enrich the training dataset, thus improving the robustness of the voice conversion process. Fine-tuning was conducted using a diverse dataset, leading to a model that is more resilient to variations in speaker characteristics.

In conclusion, the work presented in this thesis has significantly advanced the field of voice conversion for Vietnamese. The innovative approaches adopted, such as the integration of the Text2SSL module and the Spectrogram-Resize augmentation, have demonstrated promising results in overcoming the obstacles associated with tonal language conversion and the scarcity of comprehensive Vietnamese speech datasets. The proposed model not only improved upon the baseline Phoneme Hallucinator model in handling tonal nuances but also exhibited superior performance in terms of intelligibility and speaker similarity, particularly in one-shot voice conversion scenarios.

Looking forward, the techniques and findings from this research offer a solid foundation for future exploration. The improvements in model architecture and training strategies suggest a pathway towards more nuanced and sophisticated voice conversion systems that could potentially extend to other tonal languages. The success of the proposed model exemplifies the potential of targeted modifications and fine-tuning in the development of highly effective voice conversion systems that are finely attuned to the linguistic characteristics of specific languages.

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